

RPSC · RAS & RTS PRELIMINARY EXAMINATION 2026

Geography of India

230 practice questions across 6 chapters, each chapter opening with a must-know fact sheet. Answers and explanations are at the back, so you can work through the paper first.

IG01	Physiographic Divisions of India	40
IG02	Climate of India	35
IG03	Rivers, Lakes and Drainage of India	40
IG04	Irrigation and Multipurpose Projects of India	30
IG05	Agriculture and Crops of India	40
IG06	Minerals, Industries and Transport of India	45

230 questions · **176** fact-sheet points · **29** tagged as previously asked
Paper pattern: 150 questions · 200 marks · 3 hours · 1/3 negative marking

IG01

Physiographic Divisions of India

40 questions · 30 fact-sheet points

Must-know facts

1. India lies between 8°4'N-37°6'N and 68°7'E-97°25'E; area 32.87 lakh sq km; 7th largest country; about 2.4% of the world's land area.
2. Indira Point in Great Nicobar (6°45'N) is India's southernmost point; Kanniyakumari (8°4'N) is the southernmost point of the mainland.
3. Standard Meridian 82°30'E passes through Mirzapur (UP) and five states — UP, MP, Chhattisgarh, Odisha, Andhra Pradesh; IST = GMT + 5:30.
4. Tropic of Cancer crosses eight states — Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram.
5. Land frontier 15,106.7 km; longest border with Bangladesh (4,096.7 km), shortest with Afghanistan (106 km).
6. Coastline re-assessed by the National Hydrographic Office to 11,098.81 km (earlier official figure 7,516.6 km); Gujarat longest at 2,340.62 km.
7. Three geological divisions — the Peninsular block (Archaean gneiss and granite), the Himalaya (folded Tethyan sediments) and the Indo-Ganga plain (alluvium).
8. North-to-south sequence of Himalayan ranges: Karakoram – Ladakh – Zaskar – Great Himalaya (Himadri) – Lesser Himalaya (Himachal) – Shivalik.
9. Himadri averages about 6,000 m and carries all the great peaks; Himachal 3,700–4,500 m (Pir Panjal, Dhauladhar, Mahabharat); Shivalik only 900–1,100 m.
10. Duns are longitudinal valleys between the Shivalik and the Lesser Himalaya — Dehra Dun, Kotli Dun, Patli Dun; Karewas are lacustrine deposits of the Kashmir valley.
11. Kangchenjunga (8,586 m, Sikkim) — highest peak of India and third highest in the world; K2/Godwin Austen (8,611 m) lies in the Karakoram.
12. Nanda Devi (7,816 m, Chamoli, Uttarakhand) — highest peak lying entirely within Indian territory; Kamet 7,756 m.
13. Burrard's regional divisions: Punjab Himalaya (Indus–Satluj), Kumaon Himalaya (Satluj–Kali), Nepal Himalaya (Kali–Teesta), Assam Himalaya (Teesta–Dihang).
14. Purvanchal hills: Patkai Bum, Naga, Manipur, Mizo (Lushai), Barail, Mikir and the Garo-Khasi-Jaintia group.
15. Passes — Zoji La and Khardung La (Ladakh), Banihal (J&K), Rohtang, Shipki La, Bara Lacha La and Kunzum (HP), Mana, Niti and Lipulekh (Uttarakhand), Nathu La and Jelep La (Sikkim), Bomdi La, Se La, Diphu and Pangsau (Arunachal Pradesh).
16. Northern Plain belts north to south: Bhabar (porous pebbles, streams vanish) → Terai (marshy, streams re-emerge) → Bhangar (old alluvium with kankar) → Khadar (new alluvium of flood plains).
17. Punjab doabs: Bist (Beas–Satluj), Bari (Beas–Ravi), Rechna (Ravi–Chenab), Chaj (Chenab–Jhelum), Sindh Sagar (Jhelum–Indus).
18. Aravalli — oldest fold mountain system of India, running north-east to south-west from Delhi to Gujarat; highest peak Guru Shikhar (1,722 m, Mount Abu, Sirohi).
19. Deccan Trap is basaltic lava of Cretaceous-Eocene age; its weathering yields black regur (cotton) soil.
20. Malwa plateau lies between the Aravalli and the Vindhya; the Chotanagpur plateau is drained by the Damodar; the Meghalaya plateau is cut off from the main block by the Malda (Garo-Rajmahal) gap.

21. Dhupgarh (1,350 m) near Pachmarhi is the highest point of the Satpura; the Maikal range links Satpura to Chotanagpur; Amarkantak is the Narmada–Son–Johilla source area.
22. Western Ghats (Sahyadri) are continuous and crossed only by passes; the Eastern Ghats are discontinuous and dissected by the east-flowing rivers.
23. Anamudi (2,695 m, Idukki–Ernakulam, Kerala) — highest peak of the Western Ghats, of South India and of the whole of India south of the Himalaya.
24. Doddabetta (2,637 m) is the highest peak of the Nilgiri hills; Arma Konda (1,680 m, Alluri Sitharama Raju district, Andhra Pradesh) is the highest peak of the Eastern Ghats.
25. The Eastern and Western Ghats meet at the Nilgiris; the Palghat (Palakkad) gap separates the Nilgiri from the Anaimalai hills; Thal Ghat and Bhor Ghat cross the Sahyadri in Maharashtra.
26. Western coastal plain — narrow, submergent: Kachchh–Kathiawar, Gujarat, Konkan, Kanara and Malabar (backwaters called kayals).
27. Eastern coastal plain — wide, emergent and deltaic: Utkal, Northern Circars/Andhra and Coromandel; Chilika and Pulicat lie along it.
28. Andaman & Nicobar: Ten Degree Channel separates Little Andaman from Car Nicobar; Duncan Passage separates South Andaman from Little Andaman; Coco Channel lies towards Myanmar.
29. Barren Island is India's only active volcano and Narcondam is dormant; Saddle Peak (732 m, North Andaman) is the highest point of the group.
30. Lakshadweep is of coral origin with headquarters at Kavaratti; the Nine Degree Channel separates Minicoy from the main group and the Eight Degree Channel separates Minicoy from the Maldives.

1. Which mountain peak, the third highest in the world, lies on the India–Nepal border in Sikkim?

- | | |
|------------|-------------------|
| (A) Makalu | (B) Dhaulagiri |
| (C) Lhotse | (D) Kangchenjunga |

2. Which one of the following is the highest mountain peak lying entirely within Indian territory?

- | | |
|----------------|--------------------|
| (A) Kamet | (B) Trisul |
| (C) Nanda Devi | (D) Saltoro Kangri |

3. Anamudi, the highest peak of peninsular India, is located in which state? RAS PRE 2024

- | | |
|---------------|-----------------|
| (A) Kerala | (B) Tamil Nadu |
| (C) Karnataka | (D) Maharashtra |

4. Arma Konda, the highest peak of the Eastern Ghats, lies in which state?

- | | |
|---------------|--------------------|
| (A) Odisha | (B) Tamil Nadu |
| (C) Telangana | (D) Andhra Pradesh |

5. Doddabetta (2,637 m) is the highest peak of which hills? RAS PRE 2024

- | | |
|---------------------|-------------------|
| (A) Anaimalai hills | (B) Nilgiri hills |
| (C) Cardamom hills | (D) Palani hills |

6. The Eastern Ghats and the Western Ghats meet at the:

- | | |
|---------------------|--------------------|
| (A) Anaimalai hills | (B) Cardamom hills |
| (C) Nilgiri hills | (D) Palani hills |

7. India's only active volcano is situated on:

- | | |
|----------------------|--------------------|
| (A) Narcondam Island | (B) Barren Island |
| (C) Car Nicobar | (D) Little Andaman |

- 8.** The Ten Degree Channel separates:
- (A) Little Andaman from Car Nicobar (B) South Andaman from Little Andaman
(C) Minicoy from the Maldives (D) Minicoy from the main Lakshadweep group
- 9.** Indira Point, the southernmost point of India, is located in:
- (A) Little Andaman (B) Car Nicobar
(C) Great Nicobar (D) Minicoy
- 10.** The Standard Meridian of India (82°30' E) passes through which city?
- (A) Prayagraj (B) Kanpur
(C) Varanasi (D) Mirzapur
- 11.** The Tropic of Cancer passes through how many Indian states?
- (A) Six (B) Seven
(C) Eight (D) Nine
- 12.** India shares its longest land boundary with:
- (A) China (B) Bangladesh
(C) Pakistan (D) Nepal
- 13.** The narrow belt of porous pebbly deposits along the Himalayan foothills, in which the descending streams disappear underground, is called the:
- (A) Terai (B) Bhabar
(C) Bhangar (D) Khadar
- 14.** The newer alluvium deposited on the flood plains of the northern plains is known as:
- (A) Bhangar (B) Bhabar
(C) Terai (D) Khadar
- 15.** The longitudinal valleys lying between the Shivalik and the Lesser Himalaya are known as:
- (A) Duns (B) Karewas
(C) Buggals (D) Doabs
- 16.** Guru Shikhar, the highest peak of the Aravalli range, lies in which district of Rajasthan?
- (A) Sirohi (B) Pali
(C) Udaipur (D) Rajsamand
- 17.** The Deccan Trap is composed mainly of:
- (A) Granite (B) Gneiss
(C) Sandstone (D) Basalt
- 18.** The Meghalaya plateau is separated from the main block of the Peninsular Plateau by the: **HARD**
- (A) Malda gap (B) Palghat gap
(C) Shencottah gap (D) Thal Ghat
- 19.** Which Indian state has the longest coastline?
- (A) Tamil Nadu (B) Andhra Pradesh
(C) Maharashtra (D) Gujarat
- 20.** According to the re-assessment carried out by the National Hydrographic Office, the length of India's coastline is: **HARD**
- (A) 7,516.6 km (B) 6,100 km
(C) 11,098.81 km (D) 15,106.7 km

21. The Zoji La pass provides a road link between:

- (A) Srinagar and Leh
- (B) Manali and Leh
- (C) Shimla and Kaza
- (D) Gangtok and Lhasa

22. Which one of the following ranges lies immediately to the north of the Great Himalaya (Himadri)? RAS PRE 2024

- (A) Shivalik
- (B) Pir Panjal
- (C) Zaskar
- (D) Dhauladhar

23. Consider the following statements about the Shivalik range:

- I. It is the southernmost and geologically youngest range of the Himalaya.
- II. It is built largely of unconsolidated sediments brought down by rivers from the main Himalaya.
- III. Its average elevation exceeds 3,000 metres.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

24. Consider the following statements:

- I. In the Bhabar belt the streams descending from the Himalaya disappear into porous pebble beds.
- II. The Terai lies south of the Bhabar and is a marshy, damp and thickly forested belt where the streams re-emerge.

Which of the statements given above is/are correct?

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

25. Consider the following statements about the Peninsular Plateau of India:

- I. It is composed mainly of ancient crystalline, igneous and metamorphic rocks.
- II. The Deccan Trap of the north-western Deccan is of volcanic origin.
- III. The Chotanagpur plateau is drained by the Damodar river.

Which of the statements given above are correct?

- (A) I and II only
- (B) I, II and III
- (C) II and III only
- (D) I and III only

26. Consider the following statements about the Andaman and Nicobar Islands:

- I. Saddle Peak in North Andaman is the highest point of the island group.
- II. Barren Island is the only active volcano in Indian territory.
- III. The Duncan Passage separates Little Andaman from Car Nicobar.

Which of the statements given above are correct?

- (A) II and III only
- (B) I and III only
- (C) I and II only
- (D) I, II and III

27. Consider the following statements about the Lakshadweep islands:

- I. They are of coral origin.
- II. Kavaratti is the administrative headquarters.
- III. The Eight Degree Channel separates Minicoy from the Maldives.

Which of the statements given above are correct?

- (A) I, II and III
- (B) I and II only
- (C) I and III only
- (D) II and III only

28. Consider the following statements:

- I. The western coastal plain is narrow and is an example of a submerged coast.
- II. The eastern coastal plain is wider and carries the well-developed deltas of the major east-flowing rivers.
- III. The Malabar coast is noted for its lagoons known as kayals.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

29. Consider the following statements about the Aravalli range:

- I. It is one of the oldest fold mountain systems in the world.
- II. It runs in a north-east to south-west direction from Delhi to Gujarat.
- III. Its highest peak is Guru Shikhar.

Which of the statements given above are correct? RAS PRE 2023, 2024

- (A) I and II only
- (B) I, II and III
- (C) I and III only
- (D) II and III only

30. Consider the following statements about the regional divisions of the Himalaya:

- I. The Punjab Himalaya lies between the Indus and the Satluj.
- II. The Kumaon Himalaya lies between the Satluj and the Kali.
- III. The Assam Himalaya lies between the Teesta and the Dihang.

Which of the statements given above are correct? HARD

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

31. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Pass) List-II (State/UT)

- A. Shipki La 1. Sikkim
- B. Nathu La 2. Himachal Pradesh
- C. Se La 3. Uttarakhand
- D. Lipulekh 4. Arunachal Pradesh

- (A) A-2, B-4, C-1, D-3
- (B) A-2, B-1, C-4, D-3
- (C) A-3, B-1, C-4, D-2
- (D) A-2, B-1, C-3, D-4

32. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Peak) List-II (Range/Hills)

- A. Anamudi 1. Nilgiri hills
 B. Doddabetta 2. Aravalli range
 C. Guru Shikhar 3. Satpura range
 D. Dhupgarh 4. Anaimalai hills

- (A) A-4, B-1, C-2, D-3
 (B) A-1, B-4, C-2, D-3
 (C) A-4, B-1, C-3, D-2
 (D) A-2, B-1, C-4, D-3

33. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Hills) List-II (State)

- A. Cardamom hills 1. Rajasthan
 B. Javadi hills 2. Kerala
 C. Rajmahal hills 3. Tamil Nadu
 D. Mukundara hills 4. Jharkhand

- (A) A-2, B-4, C-3, D-1
 (B) A-3, B-2, C-4, D-1
 (C) A-2, B-3, C-4, D-1
 (D) A-2, B-3, C-1, D-4

34. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Water body) List-II (Separates)

- A. Ten Degree Channel 1. Minicoy and the Maldives
 B. Duncan Passage 2. Little Andaman and Car Nicobar
 C. Nine Degree Channel 3. South Andaman and Little Andaman
 D. Eight Degree Channel 4. Minicoy and the main Lakshadweep group

- (A) A-3, B-2, C-4, D-1
 (B) A-2, B-3, C-1, D-4
 (C) A-2, B-4, C-3, D-1
 (D) A-2, B-3, C-4, D-1

35. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Doab) List-II (Rivers between which it lies)

- A. Bist Doab 1. Beas and Ravi
 B. Bari Doab 2. Ravi and Chenab
 C. Rechna Doab 3. Chenab and Jhelum
 D. Chaj Doab 4. Beas and Satluj HARD

- (A) A-1, B-4, C-2, D-3
 (B) A-4, B-1, C-2, D-3
 (C) A-4, B-2, C-1, D-3
 (D) A-4, B-1, C-3, D-2

36. Which one of the following pairs is NOT correctly matched?

- (A) Kangchenjunga — Sikkim
 (B) Nanda Devi — Uttarakhand
 (C) Anamudi — Tamil Nadu
 (D) Guru Shikhar — Rajasthan

37. Which one of the following hill ranges is NOT a part of the Purvanchal?

- (A) Patkai Bum
 (B) Naga hills
 (C) Mizo hills
 (D) Kaimur hills

38. Which one of the following statements about the Northern Plains of India is NOT correct?

- (A) The Khadar belt lies away from the flood plains and is not renewed by fresh silt.
- (B) They have been built by the alluvium of the Indus, Ganga and Brahmaputra systems.
- (C) Bhangar represents the older alluvium and often contains calcareous kankar nodules.
- (D) The Bhabar belt lies immediately south of the Shivalik foothills.

39. Which one of the following pairs of a gap or ghat and its location is NOT correctly matched?

- (A) Palghat gap — between the Nilgiri and Anaimalai hills
- (B) Bhore Ghat — Karnataka
- (C) Thal Ghat — Maharashtra
- (D) Shencottah gap — Kerala-Tamil Nadu border

40. Arrange the following ranges in the correct sequence from north to south:

I. Shivalik

II. Karakoram

III. Zaskar

IV. Ladakh RAS PRE 2024

- (A) II, III, IV, I
- (B) II, IV, III, I
- (C) IV, II, III, I
- (D) II, IV, I, III

IG02

Climate of India

35 questions · 30 fact-sheet points

Must-know facts

1. India has a tropical monsoon climate; the Himalaya shuts out cold Central Asian air and the Tropic of Cancer divides the country into a tropical south and a sub-tropical north.
2. IMD's four seasons: Winter (January–February), Pre-monsoon or hot weather (March–May), South-west monsoon (June–September), Post-monsoon or retreating monsoon (October–December).
3. Western disturbances are extra-tropical cyclones that originate over the Mediterranean region and are carried to India by the subtropical westerly jet stream.
4. Winter rain from western disturbances (locally mahawat) is small in amount but vital for the rabi wheat crop of Punjab, Haryana, western UP and northern Rajasthan.
5. The subtropical westerly jet stream lies south of the Himalaya in winter; its withdrawal north of the Tibetan plateau in early June is a precondition for the monsoon burst.
6. The tropical easterly jet stream develops over peninsular India in summer above the intensely heated Tibetan plateau.
7. The Somali (Findlater) jet is a low-level cross-equatorial jet off the East African coast that strengthens the Arabian Sea branch of the south-west monsoon.
8. The ITCZ shifts north in summer as the monsoon trough; the south-east trades cross the equator, are deflected right by the Coriolis force and arrive as south-west monsoon winds.
9. Normal onset of the south-west monsoon over Kerala is 1 June; it covers the whole country by about 8 July (IMD normal dates).
10. IMD's revised normal date for the start of withdrawal from west Rajasthan is 17 September (earlier 1 September); withdrawal from the country is complete by 15 October.
11. The south-west monsoon has two branches — the Arabian Sea branch and the Bay of Bengal branch — which converge over the north-western plains.
12. The Arabian Sea branch gives heavy rain on the windward (western) slopes of the Western Ghats and leaves a rain-shadow on the Deccan plateau.
13. Mawsynram in the Khasi hills of Meghalaya records the world's highest average annual rainfall; funnel-shaped valleys trap the Bay branch.
14. West Rajasthan (Jaisalmer, Barmer) and Ladakh receive less than 20 cm of rainfall; Leh is a cold desert.
15. A 'break' in the monsoon is a spell of dry weather during the rainy season, caused by shifts in the position of the monsoon trough.
16. The retreating monsoon season brings the oppressive 'October heat' — high temperature with high humidity and clear skies.
17. The north-east or retreating monsoon gives the Coromandel (Tamil Nadu) coast its main rainfall in October–December, the winds picking up moisture over the Bay of Bengal.
18. The Tamil Nadu coast stays dry in the south-west monsoon because it lies in the rain-shadow of the Western Ghats and runs parallel to the Bay of Bengal branch.
19. Rainfall variability is inversely related to rainfall amount — above 50% in west Rajasthan, Ladakh and the interior Deccan, below 25% on the west coast, the north-east and the Himalaya.
20. Mumbai receives over 2,000 mm of rain concentrated in June–September; Chennai receives about 1,200–1,400 mm with its maximum in October–November.

21. El Nino (warm eastern Pacific) is associated with weak or deficient Indian monsoons; La Nina with stronger ones; ENSO is tracked by the Southern Oscillation Index (Tahiti minus Darwin pressure).
22. A positive Indian Ocean Dipole (warmer western Indian Ocean) favours a good monsoon and can offset the effect of an El Nino.
23. MONEX (Monsoon Experiment) was an international field programme under the Global Weather Experiment — Winter MONEX in December 1978 and Summer MONEX in May–July 1979 over the Arabian Sea and Bay of Bengal.
24. Koppen's Indian types: Amw (west coast), As (Coromandel coast), Aw (tropical savanna of the peninsula), BShw (semi-arid steppe), BWhw (hot desert, west Rajasthan), Cwg (Ganga plain), Dfc (Arunachal Pradesh), E (polar, high Himalaya).
25. Local winds and showers: Loo (northern plains, May–June), Kal Baisakhi (West Bengal), Bardoli Chheerha (Assam), Mango showers (Kerala and coastal Karnataka), Blossom showers (coffee areas of Karnataka).
26. Tropical cyclones of the Bay of Bengal are about four times as frequent as those of the Arabian Sea, with peaks before and after the monsoon; the Odisha–Andhra coast is the most vulnerable.
27. Cyclones of the North Indian Ocean are named by IMD as the Regional Specialised Meteorological Centre, New Delhi, on behalf of thirteen WMO/ESCAP member countries.
28. IMD was established in 1875 and is headquartered at New Delhi; it is the national meteorological service of India.
29. IMD drought scale: rainfall deficiency of 26–50% of normal is a moderate drought, more than 50% a severe drought.
30. The Brahmaputra valley of Assam, the Kosi belt of north Bihar and the Damodar basin of West Bengal are among India's most flood-prone regions.

1. The western disturbances that bring winter rainfall to north-western India originate over the:

(A) Bay of Bengal	(B) Arabian Sea
(C) Mediterranean Sea	(D) Caspian Sea
2. The winter rainfall of north-western India brought by western disturbances is locally known as:

(A) Loo	(B) Kal Baisakhi
(C) Bardoli Chheerha	(D) Mahawat
3. The normal date of onset of the south-west monsoon over Kerala is:

(A) 15 May	(B) 1 June
(C) 10 June	(D) 15 June
4. As per IMD's revised normal dates, the withdrawal of the south-west monsoon begins from west Rajasthan around: HARD

(A) 1 September	(B) 10 September
(C) 17 September	(D) 1 October
5. The Tamil Nadu coast receives the bulk of its annual rainfall during October–December from the: RAS PRE 2023, 2024

(A) South-west monsoon	(B) Western disturbances
(C) Retreating (north-east) monsoon	(D) Local convectional storms
6. The Coromandel coast remains comparatively dry during the south-west monsoon season mainly because: RAS PRE 2023

(A) It lies in the rain-shadow of the Western Ghats and runs parallel to the Bay of Bengal branch
(B) It lies too far from the sea
(C) The ITCZ does not shift northward over it
(D) The Eastern Ghats are too high for the monsoon winds to cross

7. Which place records the highest average annual rainfall in India?
- (A) Cherrapunji (B) Agumbe
(C) Amini Divi (D) Mawsynram
8. The 'Loo' is:
- (A) A hot, dry westerly wind of the northern plains in May and June
(B) A pre-monsoon shower of the Malabar coast
(C) A thunderstorm of West Bengal in March
(D) A cold wind of the Kashmir valley in January
9. 'Kal Baisakhi', the violent pre-monsoon thunderstorm, is associated with which state?
- (A) Kerala (B) West Bengal
(C) Punjab (D) Gujarat
10. 'Mango showers' are pre-monsoon showers that occur in:
- (A) Assam and Meghalaya (B) Kerala and coastal Karnataka
(C) Punjab and Haryana (D) Odisha and West Bengal
11. The tropical easterly jet stream over peninsular India in summer develops mainly because of the:
- (A) Cooling of the Indian Ocean (B) Southward shift of the ITCZ
(C) Melting of Himalayan glaciers (D) Intense heating of the Tibetan plateau
12. The low-level cross-equatorial jet stream that strengthens the Arabian Sea branch of the south-west monsoon is known as the: **HARD**
- (A) Somali (Findlater) jet (B) Subtropical westerly jet
(C) Tropical easterly jet (D) Polar night jet
13. El Nino conditions in the Pacific Ocean are generally associated with:
- (A) Above-normal monsoon rainfall over India
(B) Early onset of the monsoon over Kerala
(C) Increased cyclone frequency in the Arabian Sea
(D) Deficient monsoon rainfall over India
14. The Southern Oscillation Index is computed from the sea-level pressure difference between: **HARD**
- (A) Tahiti and Darwin (B) Mumbai and Chennai
(C) Jakarta and Nairobi (D) Cape Town and Perth
15. MONEX, the international field programme of 1978–79, was undertaken to study:
- (A) Antarctic ozone depletion (B) The mechanism of the monsoon
(C) El Nino in the Pacific Ocean (D) The retreat of Himalayan glaciers
16. In Koppen's classification, western Rajasthan is placed under which climatic type? **HARD**
- (A) Aw (B) BWhw
(C) BShw (D) Cwg
17. The coefficient of variability of annual rainfall in India is highest in: **RAS PRE 2024**
- (A) The western coast (B) The north-eastern states
(C) Western Rajasthan and Ladakh (D) The Ganga plain
18. The India Meteorological Department was established in the year:
- (A) 1864 (B) 1901
(C) 1947 (D) 1875

19. Cyclones forming in the North Indian Ocean are named by the Regional Specialised Meteorological Centre located at:

- (A) Pune
- (B) Kolkata
- (C) New Delhi
- (D) Chennai

20. Consider the following statements about western disturbances:

- I. They are extra-tropical cyclones originating over the Mediterranean region.
- II. They are carried into India by the subtropical westerly jet stream.
- III. The rainfall they bring is important for the rabi wheat crop.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

21. Consider the following statements about the mechanism of the south-west monsoon:

- I. The ITCZ shifts northward in summer and lies over the Ganga plain as the monsoon trough.
- II. The south-east trade winds cross the equator and are deflected to their right to become south-west winds.
- III. The withdrawal of the subtropical westerly jet from south of the Himalaya precedes the burst of the monsoon.

Which of the statements given above are correct?

- (A) I, II and III
- (B) I and II only
- (C) II and III only
- (D) I and III only

22. Consider the following statements about the retreating monsoon season in India:

- I. The withdrawal of the monsoon begins from north-western India.
- II. October and November bring rainfall to the Coromandel coast.
- III. The season is marked by high temperature with high humidity, popularly called 'October heat'.

Which of the statements given above are correct?

- (A) I and II only
- (B) I, II and III
- (C) I and III only
- (D) II and III only

23. Consider the following statements:

- I. El Nino conditions are generally associated with a weak Indian summer monsoon.
- II. A positive Indian Ocean Dipole generally favours a good monsoon over India.
- III. La Nina conditions are generally associated with above-normal monsoon rainfall over India.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I, II and III
- (D) I and III only

24. Consider the following statements:

- I. Mumbai receives the bulk of its annual rainfall between June and September.
- II. Chennai receives its maximum rainfall in October and November.
- III. The variability of rainfall is generally higher in regions of low rainfall than in regions of heavy rainfall.

Which of the statements given above are correct? RAS PRE 2024

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

25. Consider the following statements about tropical cyclones affecting India:

- I. Cyclones of the Bay of Bengal are about four times as frequent as those of the Arabian Sea.
- II. The Odisha–Andhra Pradesh coast is the most cyclone-prone stretch of the Indian coast.
- III. Cyclones over the Bay of Bengal are most frequent just before and just after the south-west monsoon season.

Which of the statements given above are correct?

- (A) I, II and III
- (B) I and II only
- (C) II and III only
- (D) I and III only

26. Consider the following statements:

- I. IMD classifies a rainfall deficiency of 10–25% of normal as a moderate drought.
- II. The Kosi is known as the 'Sorrow of Bihar' because of its recurrent floods.
- III. The Brahmaputra valley of Assam is among the most flood-prone regions of India.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

27. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Local wind/shower) List-II (Region)

- A. Loo 1. Assam
- B. Kal Baisakhi 2. Kerala and coastal Karnataka
- C. Mango showers 3. Northern plains
- D. Bardoli Chheerha 4. West Bengal

- (A) A-3, B-1, C-2, D-4
- (B) A-1, B-4, C-2, D-3
- (C) A-3, B-4, C-2, D-1
- (D) A-3, B-4, C-1, D-2

28. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Koppen's climatic type) List-II (Region of India)

- A. Amw 1. Western Rajasthan
- B. As 2. Malabar (west) coast
- C. BWhw 3. Arunachal Pradesh
- D. Dfc 4. Coromandel coast HARD

- (A) A-2, B-4, C-1, D-3
- (B) A-4, B-2, C-1, D-3
- (C) A-2, B-1, C-4, D-3
- (D) A-2, B-4, C-3, D-1

29. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Season as classified by IMD) List-II (Months)

- A. Winter 1. March to May
 B. Pre-monsoon 2. October to December
 C. South-west monsoon 3. January and February
 D. Post-monsoon 4. June to September

- (A) A-3, B-1, C-2, D-4
 (B) A-1, B-3, C-4, D-2
 (C) A-3, B-4, C-1, D-2
 (D) A-3, B-1, C-4, D-2

30. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Wind system) List-II (Characteristic)

- A. Subtropical westerly jet 1. Blows over peninsular India in summer above the heated Tibetan plateau
 B. Tropical easterly jet 2. Low-level cross-equatorial jet off the East African coast
 C. Somali jet 3. Lies south of the Himalaya in winter and shifts north in early June
 D. Monsoon trough 4. Northward-shifted position of the ITCZ over the Ganga plain in July **HARD**

- (A) A-1, B-3, C-2, D-4
 (B) A-3, B-1, C-2, D-4
 (C) A-3, B-2, C-1, D-4
 (D) A-3, B-1, C-4, D-2

31. Which one of the following pairs of a local wind or shower and its region is NOT correctly matched?

- (A) Loo — northern plains (B) Kal Baisakhi — West Bengal
 (C) Mango showers — Kerala (D) Bardoli Chheerha — Punjab

32. Which one of the following statements about the south-west monsoon is NOT correct?

- (A) The Arabian Sea branch causes heavy rainfall on the windward western side of the Western Ghats.
 (B) The Bay of Bengal branch is deflected westward along the Ganga plain by the Himalaya.
 (C) The Arabian Sea branch causes maximum rainfall over the Coromandel coast.
 (D) The two branches converge over the north-western plains of India.

33. Which one of the following statements about western disturbances is NOT correct?

- (A) They are extra-tropical cyclones that originate over the Mediterranean region.
 (B) They reach India along with the tropical easterly jet stream.
 (C) They give winter rainfall to Punjab, Haryana and western Uttar Pradesh.
 (D) The winter rain they bring is beneficial for the rabi wheat crop.

34. Which one of the following pairs of Koppen's climatic type and Indian region is NOT correctly matched? **HARD**

- (A) BWhw — western Rajasthan (B) As — Coromandel coast
 (C) Aw — Arunachal Pradesh (D) Cwg — Ganga plain

35. Assertion (A): The Tamil Nadu coast receives most of its annual rainfall during October to December.

Reason (R): During this period the north-east monsoon winds pick up moisture while crossing the Bay of Bengal.

Select the correct answer using the codes given below: **RAS PRE 2023**

- (A) Both A and R are true and R is the correct explanation of A
 (B) Both A and R are true but R is not the correct explanation of A
 (C) A is true but R is false
 (D) A is false but R is true

IG03

Rivers, Lakes and Drainage of India

40 questions · 32 fact-sheet points

Must-know facts

1. Himalayan rivers are perennial, snow- and rain-fed, antecedent in places and cut deep gorges; peninsular rivers are seasonal, rain-fed and flow in broad, shallow, graded valleys.
2. The Ganga basin is the largest river basin of India (about a quarter of the country's area); the Godavari basin is the largest peninsular basin.
3. Indus rises near the Bokhar Chu glacier in the Kailash range of Tibet and enters India in Ladakh; its Himalayan tributaries include the Zaskar, Shyok, Nubra, Hunza and Gilgit.
4. Indus Waters Treaty signed at Karachi on 19 September 1960 by Jawaharlal Nehru and Ayub Khan with the World Bank as a signatory; the Permanent Indus Commission implements it.
5. Under the treaty the eastern rivers (Ravi, Beas, Satluj) are for India and the western rivers (Indus, Jhelum, Chenab) for Pakistan; India held the treaty in abeyance in April 2025.
6. Jhelum rises from a spring at Verinag (Anantnag) and flows through Wular lake; Chenab is formed by the union of the Chandra and Bhaga at Tandi in Himachal Pradesh.
7. Ganga (2,525 km) is formed at Devprayag by the Bhagirathi (from Gaumukh, Gangotri glacier) and the Alaknanda (from Satopanth); it enters the plains at Haridwar.
8. Panch Prayag on the Alaknanda: Vishnuprayag (Dhauliganga), Nandprayag (Nandakini), Karnaprayag (Pindar), Rudraprayag (Mandakini), Devprayag (Bhagirathi).
9. Left-bank tributaries of the Ganga: Ramganga, Gomti, Ghaghara, Gandak, Burhi Gandak, Kosi, Mahananda. Right-bank: Yamuna, Son, Punpun.
10. Yamuna rises from the Yamunotri glacier on Bandarpunch; its longest tributary is the Tons; its peninsular tributaries are the Chambal, Sind, Betwa and Ken.
11. Ghaghara is the Karnali in Nepal; the Rapti and the Sarda (Kali) are its tributaries. The Kosi, 'Sorrow of Bihar', is formed by the Sun Kosi, Arun and Tamur (Sapta Kosi).
12. Damodar, the 'Sorrow of Bengal', drains the Chotanagpur plateau and joins the Hooghly; the Damodar Valley Corporation (1948) was India's first multipurpose river valley project.
13. Brahmaputra is the Tsangpo in Tibet, enters Arunachal Pradesh as the Siang/Dihang, is joined by the Dibang and Lohit, is called the Jamuna in Bangladesh and merges with the Padma to form the Meghna.
14. Brahmaputra tributaries: Subansiri (antecedent), Kameng, Manas, Sankosh, Teesta, Dhansiri and Kopili; Majuli in Assam is the largest river island of India.
15. Narmada (1,312 km) rises at Amarkantak and flows west through a rift valley between the Vindhya and the Satpura; Dhuandhar falls and the Marble Rocks are at Jabalpur.
16. Narmada tributaries: Hiran, Burhner, Banjar, Tawa (largest), Sher, Shakkar, Barna, Kolar and Orsang.
17. Tapi (724 km) rises at Multai in Betul district of Madhya Pradesh and also occupies a rift valley; its tributaries are the Purna, Girna, Panjhra, Waghur and Bori.
18. Narmada and Tapi form estuaries in the Gulf of Khambhat instead of deltas because they flow through rift valleys.
19. Mahi crosses the Tropic of Cancer twice; the Sabarmati and the Luni rise in the Aravalli; the Luni turns saline below Balotra and ends in the Rann of Kachchh.
20. Godavari (1,465 km), the 'Dakshin Ganga', rises at Trimbakeshwar near Nashik; tributaries Penganga, Wardha, Wainganga (jointly Pranhita), Indravati, Manjra, Sabari, Purna and Pravara.
21. Krishna (1,400 km) rises near Mahabaleshwar; tributaries Koyna, Bhima, Ghataprabha, Malaprabha,

Tungabhadra and Musi.

22. Kaveri (800 km) rises at Talakaveri in the Brahmagiri hills of Kodagu, Karnataka; tributaries Hemavati, Kabini, Bhavani, Noyyal and Amaravati; Shivanasamudram falls lie on it.
23. Mahanadi rises near Sihawa in Dhamtari district of Chhattisgarh and is dammed at Hirakud; Odisha's other rivers are the Subarnarekha, Brahmani and Baitarani.
24. Waterfalls: Kunchikal on the Varahi and Jog (Gersoppa) on the Sharavathi in Karnataka; Dudhsagar on the Mandovi (Goa); Chitrakote on the Indravati (Chhattisgarh); Hundru on the Subarnarekha (Jharkhand); Athirappilly on the Chalakudy (Kerala); Nohkalikai in Meghalaya.
25. Wular (Bandipora, J&K) on the Jhelum is India's largest freshwater lake; Vembanad in Kerala is the longest lake in India.
26. Chilika (Odisha) is India's largest brackish-water lagoon and contains the Nalabana Bird Sanctuary; Pulicat (Andhra Pradesh-Tamil Nadu), behind which lies Sriharikota, is the second largest.
27. Loktak (Manipur) carries floating phumdi mats and holds Keibul Lamjao, the only floating national park, home of the sangai deer.
28. Kolleru lies between the Krishna and Godavari deltas in Andhra Pradesh; Sambhar in Rajasthan is India's largest inland saline lake; Lonar in Buldhana district, Maharashtra, is a meteorite-impact crater lake in Deccan basalt.
29. Roopkund, the 'Skeleton Lake', is a glacial lake at about 5,029 m in Chamoli district of Uttarakhand, below the Trisul massif.
30. Ramsar Convention 1971 (Ramsar, Iran); India acceded in 1982; Chilika (Odisha) and Keoladeo Ghana (Rajasthan) were designated in 1981 as India's first sites; Sundarban Wetland is the largest and Renuka (HP) among the smallest; Tamil Nadu has the largest number.
31. National Perspective Plan (1980) proposes 30 links — 16 peninsular and 14 Himalayan; NWDA was set up in 1982; the Ken-Betwa link with the Daudhan dam is the first project taken up.
32. Inter-State River Water Disputes Act, 1956 governs tribunals — Kaveri, Krishna, Godavari, Narmada (award 1979), Ravi-Beas (Eradi), Mahadayi and Vansadhara.

1. The Rapti river is a tributary of the: RAS PRE 2024

- | | |
|------------|--------------|
| (A) Gandak | (B) Ghaghara |
| (C) Kosi | (D) Gomti |

2. The Teesta river is a tributary of the: RAS PRE 2021

- | | |
|-----------------|---------------|
| (A) Ganga | (B) Mahananda |
| (C) Brahmaputra | (D) Subansiri |

3. The Hiran river is a tributary of the: RAS PRE 2024

- | | |
|-------------|---------------|
| (A) Tapi | (B) Mahi |
| (C) Narmada | (D) Sabarmati |

4. The Purna river, which rises in the Ajanta hills of Chhatrapati Sambhajinagar (Aurangabad) district and meets its main stream near Purna town in Parbhani district, is a tributary of the: RAS PRE 2024

- | | |
|--------------|-------------|
| (A) Tapi | (B) Krishna |
| (C) Godavari | (D) Bhima |

5. The Ganga is formed by the confluence of the Bhagirathi and the Alaknanda at:

- | | |
|-----------------|-----------------|
| (A) Rudraprayag | (B) Karnaprayag |
| (C) Haridwar | (D) Devprayag |

- 6.** The Chenab is formed by the union of the Chandra and the Bhaga at: HARD
- (A) Tandi (B) Verinag
(C) Kishtwar (D) Akhnoor
- 7.** The Jhelum river rises from a spring at:
- (A) Verinag (B) Kokernag
(C) Achabal (D) Gulmarg
- 8.** The Indus Waters Treaty between India and Pakistan was signed in 1960 at:
- (A) New Delhi (B) Lahore
(C) Washington D.C. (D) Karachi
- 9.** Under the Indus Waters Treaty, the waters of which group of rivers were allotted to India?
- (A) Indus, Jhelum and Chenab (B) Jhelum, Chenab and Ravi
(C) Beas, Satluj and Chenab (D) Ravi, Beas and Satluj
- 10.** Which is the longest river of peninsular India?
- (A) Godavari (B) Krishna
(C) Mahanadi (D) Kaveri
- 11.** The Kaveri river rises at Talakaveri in which hills?
- (A) Brahmagiri hills (B) Nilgiri hills
(C) Anaimalai hills (D) Palani hills
- 12.** Which pair of peninsular rivers flows through rift valleys and forms estuaries instead of deltas?
- (A) Narmada and Tapi (B) Mahanadi and Godavari
(C) Krishna and Kaveri (D) Mahi and Sabarmati
- 13.** Which Indian river crosses the Tropic of Cancer twice?
- (A) Narmada (B) Luni
(C) Sabarmati (D) Mahi
- 14.** Which is the largest freshwater lake of India?
- (A) Chilika (B) Loktak
(C) Vembanad (D) Wular
- 15.** Which is the longest lake in India?
- (A) Vembanad (B) Chilika
(C) Pulicat (D) Kolleru
- 16.** Keibul Lamjao, the only floating national park in the world, is situated on which lake?
- (A) Chilika (B) Loktak
(C) Kolleru (D) Deepor Beel
- 17.** Roopkund, the high-altitude 'Skeleton Lake', is located in which state?
- (A) Himachal Pradesh (B) Sikkim
(C) Uttarakhand (D) Jammu and Kashmir
- 18.** Chilika, the largest brackish-water lagoon of India, lies in which state?
- (A) Andhra Pradesh (B) West Bengal
(C) Odisha (D) Tamil Nadu

19. Majuli, the largest river island of India, lies on which river?

- (A) Ganga (B) Godavari
(C) Mahanadi (D) Brahmaputra

20. Which two wetlands were designated as India's first Ramsar sites in 1981? RAS PRE 2016, 2023

- (A) Wular and Loktak (B) Chilika and Keoladeo Ghana
(C) Sambhar and Harike (D) Vembanad and Ashtamudi

21. The first project taken up under the National Perspective Plan for interlinking of rivers is the link between the:

- (A) Ken and the Betwa (B) Par, Tapi and Narmada
(C) Godavari and the Krishna (D) Damanganga and the Pinjal

22. Consider the following statements:

- I. Himalayan rivers are perennial because they are fed by both snowmelt and rainfall.
II. Peninsular rivers generally flow in broad, shallow valleys with a nearly graded profile.
III. The Himalayan rivers build large deltas where they reach the sea.

Which of the statements given above are correct?

- (A) I and II only
(B) I, II and III
(C) II and III only
(D) I and III only

23. Consider the following statements about the Indus Waters Treaty:

- I. It was signed in 1960 with the World Bank as a signatory.
II. The waters of the Ravi, Beas and Satluj were allotted to India.
III. A Permanent Indus Commission was constituted under the treaty.

Which of the statements given above are correct?

- (A) I and II only
(B) II and III only
(C) I, II and III
(D) I and III only

24. Consider the following statements about the Brahmaputra:

- I. It is known as the Tsangpo in Tibet.
II. It enters India in Arunachal Pradesh as the Siang or Dihang.
III. It is called the Padma in Bangladesh.

Which of the statements given above are correct?

- (A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

25. Consider the following statements about the Narmada:

- I. It rises at Amarkantak in Madhya Pradesh.
II. It flows westward through a rift valley between the Vindhya and the Satpura.
III. It forms the Dhuandhar falls near Jabalpur.

Which of the statements given above are correct?

- (A) II and III only
(B) I and III only
(C) I, II and III
(D) I and II only

26. Consider the following statements about Chilika lake:

- I. It is the largest brackish-water lagoon in India.
- II. It was designated a Ramsar site in 1981.
- III. The Nalabana Bird Sanctuary lies within it.

Which of the statements given above are correct?

- (A) I and II only
- (B) I, II and III
- (C) I and III only
- (D) II and III only

27. Consider the following statements:

- I. Lonar is a meteorite-impact crater lake formed in the basalt of the Deccan Trap in Maharashtra.
- II. Sambhar is the largest inland saline lake of India.
- III. Pulicat is the second largest brackish-water lagoon of India.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

28. Consider the following statements about the interlinking of rivers in India:

- I. The National Perspective Plan of 1980 envisages 30 links, 16 peninsular and 14 Himalayan.
- II. The National Water Development Agency was set up in 1982.
- III. The Ken–Betwa link involves the construction of the Daudhan dam.

Which of the statements given above are correct? **HARD**

- (A) I, II and III
- (B) I and II only
- (C) II and III only
- (D) I and III only

29. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Tributary) List-II (Main river)

- A. Rapti 1. Brahmaputra
- B. Teesta 2. Godavari
- C. Hiran 3. Ghaghara
- D. Purna 4. Narmada

RAS PRE 2021, 2024

- (A) A-3, B-1, C-2, D-4
- (B) A-3, B-1, C-4, D-2
- (C) A-1, B-3, C-4, D-2
- (D) A-3, B-4, C-1, D-2

30. Match List-I with List-II and select the correct answer using the codes given below:

List-I (River) List-II (Place of origin)

- A. Godavari 1. Multai (Betul)
- B. Krishna 2. Trimbakeshwar (Nashik)
- C. Mahanadi 3. Mahabaleshwar
- D. Tapi 4. Sihawa (Dhamtari)

- (A) A-2, B-3, C-1, D-4
- (B) A-3, B-2, C-4, D-1
- (C) A-2, B-3, C-4, D-1
- (D) A-2, B-4, C-3, D-1

31. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Lake) List-II (State/UT)

- A. Kolleru 1. Manipur
 B. Loktak 2. Andhra Pradesh
 C. Lonar 3. Uttarakhand
 D. Roopkund 4. Maharashtra

- (A) A-1, B-2, C-4, D-3
 (B) A-2, B-1, C-4, D-3
 (C) A-2, B-1, C-3, D-4
 (D) A-2, B-4, C-1, D-3

32. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Waterfall) List-II (River)

- A. Jog 1. Indravati
 B. Dudhsagar 2. Sharavathi
 C. Chitrakote 3. Subarnarekha
 D. Hundru 4. Mandovi

- (A) A-2, B-4, C-3, D-1
 (B) A-4, B-2, C-1, D-3
 (C) A-2, B-1, C-4, D-3
 (D) A-2, B-4, C-1, D-3

33. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Confluence) List-II (River meeting the Alaknanda there)

- A. Vishnuprayag 1. Pindar
 B. Nandprayag 2. Mandakini
 C. Karnaprayag 3. Dhauliganga
 D. Rudraprayag 4. Nandakini

HARD

- (A) A-3, B-1, C-4, D-2
 (B) A-3, B-4, C-2, D-1
 (C) A-3, B-4, C-1, D-2
 (D) A-4, B-3, C-1, D-2

34. Which one of the following pairs of a river and its tributary is NOT correctly matched?

- (A) Ganga — Gomti (B) Krishna — Sabari
 (C) Yamuna — Betwa (D) Godavari — Indravati

35. Which one of the following pairs of a lake and its state is NOT correctly matched?

- (A) Kanwar (Kabartal) — West Bengal (B) Wular — Jammu and Kashmir
 (C) Sasthamkotta — Kerala (D) Deepor Beel — Assam

36. Which one of the following is NOT a left-bank tributary of the Ganga?

- (A) Ghaghara (B) Gandak
 (C) Kosi (D) Son

37. Which one of the following statements about the peninsular rivers of India is NOT correct?

- (A) Most of them rise in the Western Ghats and flow eastward into the Bay of Bengal.
 (B) They generally have broad and shallow valleys with a nearly graded profile.
 (C) The Narmada and the Tapi build large deltas at their mouths.
 (D) They are largely seasonal and depend on rainfall.

38. Which one of the following pairs of a waterfall and its state is NOT correctly matched?

- (A) Athirappilly — Kerala
(B) Chitrakote — Odisha
(C) Nohkalikai — Meghalaya
(D) Hundru — Jharkhand

39. Arrange the following rivers in descending order of their length:

I. Godavari

II. Ganga

III. Narmada

IV. Krishna **HARD**

- (A) II, I, IV, III
(B) II, I, III, IV
(C) I, II, IV, III
(D) II, IV, I, III

40. Arrange the following confluences on the Alaknanda in order from upstream to downstream:

I. Karnaprayag

II. Devprayag

III. Vishnuprayag

IV. Rudraprayag **HARD**

- (A) III, IV, I, II
(B) I, III, IV, II
(C) III, I, II, IV
(D) III, I, IV, II

IG04

Irrigation and Multipurpose Projects of India

30 questions · 28 fact-sheet points

Must-know facts

1. Sources of irrigation (net irrigated area): tube wells ~45%, canals ~25%, other wells ~16%, other sources ~11%, tanks ~3% — wells+tube wells together irrigate over 60%.
2. Uttar Pradesh has the largest net irrigated area of any state; Punjab has the highest share of its net sown area irrigated (about 98%).
3. Rice takes the largest share of India's irrigated area, followed by wheat; canal irrigation dominates the northern plains, tank irrigation the peninsular plateau.
4. Bhakra Dam — Sutlej — Bilaspur (Himachal Pradesh); 226 m, highest straight gravity dam in India; reservoir Gobind Sagar; Nangal dam lies downstream.
5. Bhakra-Nangal is a joint venture of Punjab, Haryana, Rajasthan and Himachal Pradesh; Nangal Hydel Channel feeds the Bhakra main line.
6. Tehri Dam — Bhagirathi and Bhilangana — Uttarakhand; 260.5 m, the tallest dam in India.
7. Hirakud Dam — Mahanadi — Odisha (1957); with its dykes it is among the longest dams in the world (about 25.8 km).
8. Damodar Valley Corporation (1948) — first multipurpose river valley project of independent India, modelled on the Tennessee Valley Authority; dams Tilaiya and Maithon (Barakar), Konar, Panchet (Damodar); Jharkhand and West Bengal.
9. Nagarjuna Sagar — Krishna — Telangana/Andhra Pradesh; one of the largest masonry dams; canals Jawahar (right bank) and Lal Bahadur (left bank).
10. Rihand Dam — Rihand river (Son tributary), Sonbhadra, Uttar Pradesh; its reservoir, Govind Ballabh Pant Sagar, is one of the largest artificial lakes in India.
11. Sardar Sarovar — Narmada — Gujarat; benefits Gujarat, Madhya Pradesh, Maharashtra and Rajasthan; Narmada main canal is among the world's largest irrigation canals.
12. Farakka Barrage (1975) — Ganga, West Bengal; diverts water into the Bhagirathi-Hooghly to flush silt and keep Kolkata port navigable.
13. Indira Gandhi Canal (formerly Rajasthan Canal) — off-takes from Harike barrage below the Sutlej-Beas confluence; Rajasthan feeder 204 km + main canal 445 km; longest canal in India.
14. Kosi project — joint Indo-Nepal venture, tames the 'Sorrow of Bihar'; Gandak project — Bihar, Uttar Pradesh and Nepal.
15. Chambal Valley project — joint Madhya Pradesh-Rajasthan: Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar and Kota barrage.
16. Tungabhadra project — Karnataka and Andhra Pradesh; Parambikulam-Aliyar — Tamil Nadu and Kerala; Bansagar (Son) — Madhya Pradesh, Uttar Pradesh and Bihar.
17. Other key pairs: Ukai-Tapi (Gujarat), Koyna-Koyna (Maharashtra), Idukki-Periyar arch dam (Kerala), Mettur/Stanley-Kaveri (Tamil Nadu), Sriramsagar-Godavari (Telangana), Nathpa Jhakri-Satluj (Himachal), Ranjit Sagar (Thein)-Ravi, Pong (Maharana Pratap Sagar)-Beas.
18. PMKSY launched 1 July 2015 with the motto 'Har Khet Ko Pani'; 2021-26 phase outlay Rs 93,068.56 crore; components — AIBP, Har Khet Ko Pani and Watershed Development.
19. 'Per Drop More Crop' (micro-irrigation: drip and sprinkler) was a PMKSY component till 2021-22; from 2022-23 it is implemented under the Rashtriya Krishi Vikas Yojana; Micro Irrigation Fund is with NABARD.

20. Command Area Development Programme launched 1974-75; now Command Area Development and Water Management under PMKSY.
21. Atal Bhujal Yojana — launched 25 December 2019 (Vajpayee's birth anniversary), Ministry of Jal Shakti; Rs 6,000 crore, 50% World Bank loan and 50% Central share; demand-side, community-led groundwater management.
22. Atal Bhujal Yojana covers seven water-stressed states — Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh; water security plans are made by gram panchayats.
23. Dynamic Ground Water Resource Assessment 2024 (CGWB): annual recharge 446.90 BCM, extractable resource 406.19 BCM, extraction 245.64 BCM, stage of extraction 60.47%.
24. Of 6,746 assessment units in 2024, 4,951 are 'safe' (73.4%), 751 'over-exploited' (11.1%), 206 'critical' and 711 'semi-critical'.
25. Traditional water harvesting: khadin and tanka (Rajasthan), johad (Alwar), kuhl (Himachal), zabo (Nagaland), apatani (Arunachal), ahar-pyne (Bihar), eri tanks (Tamil Nadu), surangam (Kerala), phad (Maharashtra), bamboo drip (Meghalaya).
26. National Perspective Plan (1980) is executed by the National Water Development Agency (1982); the Ken-Betwa Link (Daudhan dam on the Ken, Panna) is the first river-link project taken up, in Madhya Pradesh and Uttar Pradesh.
27. National Water Policy first framed in 1987 (revised 2002 and 2012); the National Water Mission under NAPCC targets a 20% improvement in water-use efficiency.
28. Jal Jeevan Mission (2019) aims at functional household tap connections ('Har Ghar Jal'); Namami Gange is the umbrella Ganga rejuvenation programme.

1. The Bhakra Dam, the highest straight gravity dam in India, is built on which river?

(A) Beas	(B) Ravi
(C) Sutlej	(D) Chenab
2. Tehri Dam, the tallest dam in India, is built at the confluence of the Bhagirathi with which river?

(A) Alaknanda	(B) Mandakini
(C) Dhauliganga	(D) Bhilangana
3. The Hirakud Dam, one of the longest dams in the world, is built on the Mahanadi in which state?

(A) Chhattisgarh	(B) Jharkhand
(C) Odisha	(D) West Bengal
4. Which was the first multipurpose river valley project of independent India, modelled on the Tennessee Valley Authority?

(A) Bhakra-Nangal Project	(B) Damodar Valley Corporation
(C) Hirakud Project	(D) Kosi Project
5. Nagarjuna Sagar, one of the largest masonry dams in the world, is built across which river?

(A) Godavari	(B) Tungabhadra
(C) Krishna	(D) Penna
6. Govind Ballabh Pant Sagar, one of India's largest artificial lakes, is the reservoir of which dam?

(A) Rihand	(B) Bansagar
(C) Gandhi Sagar	(D) Ukai

- 7.** The Farakka Barrage on the Ganga was built primarily to
- (A) generate hydro-electricity for Bihar
 - (B) divert water into the Bhagirathi-Hooghly to keep Kolkata port navigable
 - (C) control floods of the Kosi river
 - (D) supply irrigation water to the Damodar valley
- 8.** The Indira Gandhi Canal, the longest canal in India, takes off from which barrage? RAS PRE 2023
- (A) Harike
 - (B) Ferozepur
 - (C) Pong
 - (D) Ropar
- 9.** Which of the following is the largest source of irrigation in India in terms of net irrigated area?
- (A) Canals
 - (B) Tanks
 - (C) Wells and tube wells
 - (D) Other sources
- 10.** Which crop accounts for the largest share of India's irrigated area?
- (A) Wheat
 - (B) Sugarcane
 - (C) Rice
 - (D) Cotton
- 11.** The Atal Bhujal Yojana, launched on 25 December 2019, is implemented by which ministry? RAS PRE 2021
- (A) Ministry of Agriculture and Farmers' Welfare
 - (B) Ministry of Jal Shakti
 - (C) Ministry of Rural Development
 - (D) Ministry of Environment, Forest and Climate Change
- 12.** Under the Pradhan Mantri Krishi Sinchayee Yojana, which component promotes drip and sprinkler (micro) irrigation?
- (A) Har Khet Ko Pani
 - (B) Per Drop More Crop
 - (C) Accelerated Irrigation Benefit Programme
 - (D) Watershed Development
- 13.** The Command Area Development Programme was launched in which year? HARD
- (A) 1966-67
 - (B) 1980-81
 - (C) 1991-92
 - (D) 1974-75
- 14.** Besides Gujarat, Madhya Pradesh and Maharashtra, which state is a beneficiary of the Sardar Sarovar Project?
- (A) Chhattisgarh
 - (B) Uttar Pradesh
 - (C) Karnataka
 - (D) Rajasthan
- 15.** The Tungabhadra multipurpose project is a joint undertaking of which states?
- (A) Karnataka and Andhra Pradesh
 - (B) Maharashtra and Karnataka
 - (C) Odisha and Chhattisgarh
 - (D) Kerala and Tamil Nadu
- 16.** The Ken-Betwa Link Project, the first river-linking project taken up under the National Perspective Plan, involves which two states?
- (A) Madhya Pradesh and Uttar Pradesh
 - (B) Rajasthan and Madhya Pradesh
 - (C) Uttar Pradesh and Bihar
 - (D) Maharashtra and Madhya Pradesh

17. Consider the following statements about the Damodar Valley Corporation:

- I. It was set up in 1948 on the model of the Tennessee Valley Authority.
- II. Maithon and Panchet are among its dams.
- III. It serves the states of Jharkhand and West Bengal.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

18. Consider the following statements about the Atal Bhujal Yojana:

- I. Half of its outlay is met by a World Bank loan.
- II. It is being implemented in all the states of India.
- III. It stresses demand-side management of groundwater through community participation.

Which of the statements given above are correct? RAS PRE 2023

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

19. Consider the following statements about the Indira Gandhi Canal:

- I. It off-takes from the Harike barrage.
- II. It carries the waters of the Sutlej and the Beas.
- III. Its main canal terminates at Sriganganagar.

Which of the statements given above are correct?

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

20. Consider the following statements about the Dynamic Ground Water Resource Assessment, 2024:

- I. The average stage of groundwater extraction for the country is about 60 per cent.
- II. Less than half of the assessment units fall in the 'safe' category.
- III. The assessment is carried out by the Central Ground Water Board along with the states.

Which of the statements given above are correct? HARD

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

21. Consider the following statements about irrigation in India:

- I. Tanks account for nearly one-fourth of India's net irrigated area.
- II. Canal irrigation is dominant in the northern plains.
- III. Punjab has the highest proportion of its net sown area under irrigation among the major states.

Which of the statements given above are correct?

- (A) I and III only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

22. Consider the following statements about the Pradhan Mantri Krishi Sinchayee Yojana:

- I. It was launched in 2015 with the motto 'Har Khet Ko Pani'.
- II. The Accelerated Irrigation Benefit Programme is one of its components.
- III. From 2022-23 'Per Drop More Crop' is implemented under the Rashtriya Krishi Vikas Yojana.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

23. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Project) List-II (River)

- A. Hirakud 1. Krishna
- B. Nagarjuna Sagar 2. Mahanadi
- C. Bhakra 3. Chambal
- D. Gandhi Sagar 4. Sutlej

- (A) A-2, B-1, C-4, D-3
- (B) A-2, B-4, C-1, D-3
- (C) A-1, B-2, C-3, D-4
- (D) A-2, B-1, C-3, D-4

24. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Dam) List-II (State)

- A. Ukai 1. Gujarat
- B. Koyna 2. Maharashtra
- C. Idukki 3. Kerala
- D. Mettur 4. Tamil Nadu

- (A) A-2, B-1, C-4, D-3
- (B) A-1, B-2, C-4, D-3
- (C) A-1, B-2, C-3, D-4
- (D) A-3, B-2, C-1, D-4

25. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Traditional water harvesting structure) List-II (State)

- A. Khadin 1. Rajasthan
- B. Zabo 2. Nagaland
- C. Ahar-pyne 3. Bihar
- D. Kuhl 4. Himachal Pradesh

RAS PRE 2024

- (A) A-1, B-2, C-4, D-3
- (B) A-1, B-2, C-3, D-4
- (C) A-2, B-1, C-3, D-4
- (D) A-1, B-3, C-2, D-4

26. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Project) List-II (Participating states)

- A. Kosi 1. Bihar and Nepal
B. Gandak 2. Bihar, Uttar Pradesh and Nepal
C. Chambal 3. Madhya Pradesh and Rajasthan
D. Parambikulam-Aliyar 4. Tamil Nadu and Kerala **HARD**

- (A) A-2, B-1, C-3, D-4
(B) A-1, B-2, C-4, D-3
(C) A-3, B-2, C-1, D-4
(D) A-1, B-2, C-3, D-4

27. Which one of the following project-river pairs is NOT correctly matched?

- (A) Hirakud — Mahanadi (B) Bhakra — Beas
(C) Sardar Sarovar — Narmada (D) Tehri — Bhagirathi

28. Which one of the following is NOT a component of the Pradhan Mantri Krishi Sinchayee Yojana in its 2021-26 phase?

- (A) Accelerated Irrigation Benefit Programme (B) Har Khet Ko Pani
(C) Watershed Development (D) Pradhan Mantri Fasal Bima Yojana

29. Which one of the following states is NOT covered under the Atal Bhujal Yojana?

- (A) Punjab (B) Haryana
(C) Rajasthan (D) Karnataka

30. Arrange the following in the chronological order of their establishment or commissioning:

- I. Damodar Valley Corporation
II. Hirakud Dam
III. Bhakra Dam
IV. Farakka Barrage **HARD**

- (A) I, II, III, IV
(B) II, I, III, IV
(C) I, III, II, IV
(D) I, II, IV, III

IG05

Agriculture and Crops of India

40 questions · 28 fact-sheet points

Must-know facts

1. Kharif crops are sown with the onset of the south-west monsoon (June-July) and harvested in September-October: rice, jowar, bajra, maize, cotton, jute, groundnut, tur, soybean.
2. Rabi crops are sown in October-December and harvested in April-June: wheat, barley, gram, mustard, peas, linseed. Zaid is the short summer season: watermelon, muskmelon, cucumber, moong, fodder.
3. Leading producer states: wheat and sugarcane — Uttar Pradesh; jute — West Bengal; tea — Assam; coffee — Karnataka; rubber — Kerala; cotton — Gujarat (largest area under cotton is in Maharashtra).
4. Leading producer states: rapeseed-mustard, bajra and barley — Rajasthan; soybean, gram and total pulses — Madhya Pradesh; groundnut — Gujarat; jowar — Maharashtra; onion — Maharashtra; potato and mango — Uttar Pradesh.
5. India is the world's largest producer of milk, pulses, jute and spices, and the second largest producer of rice, wheat, sugarcane, groundnut, fruits and vegetables.
6. Rice needs over 100 cm rainfall and 25 degrees C; West Bengal grows three crops — aus, aman and boro; Punjab has the highest rice yield per hectare among major states.
7. Wheat is a rabi crop of 50-75 cm rainfall and a cool growing season with bright sunshine at ripening; Punjab has the highest wheat productivity.
8. Sugarcane belt: Uttar Pradesh leads in area and production; sugar recovery is higher in the tropical belt of Maharashtra and Karnataka; sugarcane gets a Fair and Remunerative Price (FRP), not MSP.
9. Jute, the 'golden fibre', needs 150-200 cm rainfall and grows in the Ganga-Brahmaputra delta; the first jute mill was set up at Rishra (1855).
10. Plantation crops: tea introduced in Assam in the 1830s (Darjeeling, Nilgiris); coffee (Arabica and Robusta) began in the Baba Budan hills of Karnataka; rubber needs over 200 cm rainfall.
11. Soils of India: alluvial (largest area, northern plains and deltas), black/regur, red and yellow, laterite, arid, forest-mountain, saline-alkaline and peaty.
12. Black (regur) soil forms from the basaltic lava of the Deccan trap; it is rich in lime, iron, magnesia and potash but poor in nitrogen, phosphorus and humus, and self-ploughs by cracking in summer.
13. Laterite soil is the product of intense leaching under high rainfall and high temperature; poor in fertility but good for tea, coffee, cashew and rubber; used as building brick.
14. Red soils form from crystalline igneous rocks; they are reddish from iron oxide and deficient in nitrogen, phosphorus and humus.
15. The Planning Commission divided India into 15 agro-climatic zones; ICAR/NARP recognises 127 agro-climatic zones and NBSS&LUP 20 agro-ecological regions.
16. Green Revolution began in 1966-67 with HYV wheat (Mexican dwarf varieties of Norman Borlaug); M. S. Swaminathan is called the father of India's Green Revolution; Punjab, Haryana and western Uttar Pradesh gained most; IR-8 was the miracle rice of IRRI.
17. White Revolution — Operation Flood (1970), NDDB, Verghese Kurien, Anand pattern cooperatives; Yellow Revolution — oilseeds, Technology Mission on Oilseeds (1986); Blue — fisheries; Silver — eggs/poultry; Round — potato; Pink — prawns; Grey — fertilisers.
18. Minimum Support Price is recommended by the Commission for Agricultural Costs and Prices (set up 1965) and announced by the CCEA for 22 crops, plus FRP for sugarcane; since 2018-19 MSP is at least 1.5 times the A2+FL cost.

19. Food Corporation of India (1965) is the central procurement, storage and distribution agency; buffer stocking norms are fixed by the government.
20. e-NAM, the electronic National Agriculture Market, was launched on 14 April 2016 and is run by the Small Farmers' Agribusiness Consortium (SFAC), linking APMC mandis online.
21. PM-KISAN (announced 24 February 2019, effective from 1 December 2018) pays Rs 6,000 a year in three equal instalments of Rs 2,000.
22. Pradhan Mantri Fasal Bima Yojana (kharif 2016) caps farmer premium at 2% for kharif, 1.5% for rabi food and oilseed crops and 5% for commercial and horticultural crops; it replaced NAIS and MNAIS.
23. Soil Health Card Scheme was launched on 19 February 2015 from Suratgarh in Rajasthan; Paramparagat Krishi Vikas Yojana promotes organic farming; Sikkim became India's first fully organic state in 2016.
24. National Food Security Mission started in 2007-08; Rashtriya Krishi Vikas Yojana in 2007; Kisan Credit Card in 1998; Operation Greens (2018) covers tomato, onion and potato (TOP).
25. Shifting cultivation is called jhum in the north-east, podu in Andhra Pradesh and Odisha, bewar and dahiya in Madhya Pradesh, kumari in Kerala and walra among the Bhils of Rajasthan.
26. Net sown area is about 46% of the reporting area — the largest single land use in India; cropping intensity is the ratio of gross cropped area to net sown area.
27. Crop research institutes: rice — Cuttack (ICAR-NRRI), sugarcane — Lucknow (IISR), potato — Shimla (CPRI), pulses — Kanpur (IIPR), wheat and barley — Karnal (IIWBR), arid horticulture — Bikaner, seed spices — Ajmer, camel — Bikaner.
28. Agriculture and allied sectors contribute roughly 18% of gross value added but support nearly half the workforce; the National Agriculture Policy, 2000 set a 4% annual growth target.

1. Which one of the following is a rabi crop?

- | | |
|-------------|------------|
| (A) Bajra | (B) Cotton |
| (C) Mustard | (D) Jute |

2. West Bengal accounts for the largest share of India's production of which crop, known as the 'golden fibre'?

- | | |
|------------|----------|
| (A) Cotton | (B) Silk |
| (C) Coir | (D) Jute |

3. Which state is the largest producer of tea in India?

- | | |
|-----------------|----------------|
| (A) West Bengal | (B) Kerala |
| (C) Assam | (D) Tamil Nadu |

4. Karnataka accounts for the bulk of India's production of which plantation crop?

- | | |
|--------------|------------|
| (A) Tea | (B) Rubber |
| (C) Cardamom | (D) Coffee |

5. Which state is the largest producer of natural rubber in India?

- | | |
|----------------|---------------|
| (A) Kerala | (B) Karnataka |
| (C) Tamil Nadu | (D) Tripura |

6. Uttar Pradesh is the leading producing state of which of the following crops?

- | | |
|---------------|-------------|
| (A) Sugarcane | (B) Cotton |
| (C) Groundnut | (D) Soybean |

- 7.** Which state is the largest producer of rapeseed and mustard in India?
- (A) Haryana (B) Madhya Pradesh
(C) Uttar Pradesh (D) Rajasthan
- 8.** Madhya Pradesh is India's leading producer of which oilseed?
- (A) Soybean (B) Groundnut
(C) Sunflower (D) Sesamum
- 9.** Which state is the largest producer of raw cotton in India?
- (A) Maharashtra (B) Gujarat
(C) Telangana (D) Punjab
- 10.** Which state is the largest producer of total pulses in India?
- (A) Rajasthan (B) Maharashtra
(C) Madhya Pradesh (D) Uttar Pradesh
- 11.** Which state is the largest producer of bajra (pearl millet) in India?
- (A) Gujarat (B) Rajasthan
(C) Maharashtra (D) Haryana
- 12.** Which soil type covers the largest area in India?
- (A) Black soil (B) Red soil
(C) Alluvial soil (D) Laterite soil
- 13.** Regur or black cotton soil is formed by the weathering of
- (A) granite and gneiss (B) basaltic lava of the Deccan trap
(C) sandstone (D) limestone
- 14.** Laterite soil is formed mainly by which process?
- (A) Deposition by rivers
(B) Deposition by wind
(C) Intense leaching under high rainfall and high temperature
(D) Glacial abrasion
- 15.** Who is regarded as the father of India's Green Revolution?
- (A) Norman Borlaug (B) Verghese Kurien
(C) M. S. Swaminathan (D) Hiralal Chaudhuri
- 16.** Operation Flood, launched in 1970, is associated with which product?
- (A) Oilseeds (B) Milk
(C) Fish (D) Eggs
- 17.** Into how many agro-climatic zones did the Planning Commission divide India? HARD
- (A) 8 (B) 15
(C) 20 (D) 127
- 18.** Minimum Support Prices for crops are recommended by
- (A) NITI Aayog
(B) the Food Corporation of India
(C) the Commission for Agricultural Costs and Prices
(D) the Ministry of Consumer Affairs, Food and Public Distribution

- 19.** The e-NAM (National Agriculture Market) electronic trading portal was launched in which year?
- (A) 2014 (B) 2016
(C) 2018 (D) 2020
- 20.** Under the Pradhan Mantri Fasal Bima Yojana, the maximum premium payable by a farmer for rabi food and oilseed crops is
- (A) 1.5 per cent of the sum insured (B) 2 per cent of the sum insured
(C) 5 per cent of the sum insured (D) 10 per cent of the sum insured
- 21.** Under PM-KISAN, how much income support does an eligible farmer family receive in a year?
- (A) Rs 4,000 (B) Rs 6,000
(C) Rs 8,000 (D) Rs 10,000
- 22.** The Soil Health Card Scheme was launched in February 2015 from which place?
- (A) Anand, Gujarat (B) Karnal, Haryana
(C) Suratgarh, Rajasthan (D) Ludhiana, Punjab
- 23.** Which state became India's first fully organic state?
- (A) Himachal Pradesh (B) Sikkim
(C) Uttarakhand (D) Meghalaya
- 24.** Consider the following statements about rice cultivation in India:
- I. Rice is a kharif crop requiring high temperature and rainfall above 100 cm.
II. In West Bengal three crops of rice — aus, aman and boro — are raised.
III. Punjab has the highest yield of rice per hectare among the major states.
- Which of the statements given above are correct?
- (A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III
- 25.** Consider the following statements about the Green Revolution in India:
- I. It began with the introduction of high-yielding varieties of wheat in the mid-1960s.
II. Punjab, Haryana and western Uttar Pradesh were its main beneficiaries.
III. It was based on IR-8, a dwarf wheat variety developed in Mexico.
- Which of the statements given above are correct?
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III
- 26.** Consider the following statements about black soil:
- I. It is also known as regur soil.
II. It is rich in nitrogen, phosphorus and organic matter.
III. It develops deep cracks in summer which help aeration of the soil.
- Which of the statements given above are correct?
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

27. Consider the following statements about the Minimum Support Price system:

- I. MSP is announced for 22 crops besides the Fair and Remunerative Price for sugarcane.
- II. Since 2018-19 MSP has been fixed at not less than 1.5 times the A2+FL cost of production.
- III. The Food Corporation of India is a principal central procurement agency.

Which of the statements given above are correct?

- (A) II and III only
- (B) I, II and III
- (C) I and II only
- (D) I and III only

28. Consider the following statements about the Pradhan Mantri Fasal Bima Yojana:

- I. It replaced the National Agricultural Insurance Scheme and the Modified NAIS.
- II. It has been made voluntary for loanee farmers.
- III. The farmer's premium for commercial and horticultural crops is capped at 5 per cent.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

29. Consider the following statements about India's position in world agriculture:

- I. India is the world's largest producer of milk.
- II. India is the world's largest producer of pulses and jute.
- III. India is the world's largest producer of wheat.

Which of the statements given above are correct? HARD

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

30. Consider the following statements about the zaid season:

- I. Zaid crops are grown in the short season between the rabi harvest and the kharif sowing.
- II. Watermelon, muskmelon and cucumber are zaid crops.
- III. Gram and mustard are typical zaid crops.

Which of the statements given above are correct?

- (A) I and III only
- (B) II and III only
- (C) I and II only
- (D) I, II and III

31. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Crop) List-II (Leading producer state)

- A. Jowar 1. Maharashtra
- B. Wheat 2. Uttar Pradesh
- C. Groundnut 3. Gujarat
- D. Mulberry silk 4. Karnataka

- (A) A-1, B-2, C-3, D-4
- (B) A-1, B-3, C-2, D-4
- (C) A-2, B-1, C-3, D-4
- (D) A-1, B-2, C-4, D-3

32. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Revolution) List-II (Product)

- A. Yellow Revolution 1. Oilseeds
 B. Blue Revolution 2. Fisheries
 C. Silver Revolution 3. Eggs and poultry
 D. Round Revolution 4. Potato

- (A) A-2, B-1, C-3, D-4
 (B) A-1, B-2, C-4, D-3
 (C) A-1, B-2, C-3, D-4
 (D) A-3, B-2, C-1, D-4

33. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Soil) List-II (Typical region)

- A. Laterite soil 1. Malabar coast and Western Ghats
 B. Black soil 2. Deccan plateau
 C. Alluvial soil 3. Northern plains
 D. Arid soil 4. Western Rajasthan

- (A) A-2, B-1, C-3, D-4
 (B) A-1, B-2, C-4, D-3
 (C) A-3, B-2, C-1, D-4
 (D) A-1, B-2, C-3, D-4

34. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Name of shifting cultivation) List-II (Region)

- A. Jhum 1. North-eastern India
 B. Podu 2. Andhra Pradesh and Odisha
 C. Walra 3. Rajasthan
 D. Kumari 4. Kerala

RAS PRE 2013, 2018

- (A) A-1, B-3, C-2, D-4
 (B) A-1, B-2, C-3, D-4
 (C) A-2, B-1, C-3, D-4
 (D) A-1, B-2, C-4, D-3

35. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Research institute) List-II (Headquarters)

- A. National Rice Research Institute 1. Cuttack
 B. Indian Institute of Sugarcane Research 2. Lucknow
 C. Central Potato Research Institute 3. Shimla
 D. Indian Institute of Pulses Research 4. Kanpur

HARD

- (A) A-2, B-1, C-3, D-4
 (B) A-1, B-2, C-4, D-3
 (C) A-1, B-2, C-3, D-4
 (D) A-3, B-2, C-1, D-4

36. Which one of the following crop-state pairs is NOT correctly matched?

- (A) Soybean — Madhya Pradesh (B) Rubber — Kerala
 (C) Saffron — Jammu and Kashmir (D) Coffee — Assam

37. Which one of the following is NOT a kharif crop?

- (A) Jowar (B) Bajra
 (C) Barley (D) Cotton

38. Which one of the following statements about laterite soil is NOT correct?

- (A) It forms under high rainfall with alternating wet and dry periods
- (B) It is rich in humus and nitrogen
- (C) It is suited to tea, coffee and cashew cultivation
- (D) It is rich in iron and aluminium oxides

39. Which one of the following scheme-year pairs is NOT correctly matched?

- | | |
|------------------------------------|---|
| (A) e-NAM — 2016 | (B) PM-KISAN — 2019 |
| (C) Soil Health Card Scheme — 2015 | (D) National Food Security Mission — 2016 |

40. Arrange the following in the chronological order of their launch:

- I. Operation Flood
- II. National Bank for Agriculture and Rural Development
- III. National Food Security Mission
- IV. Pradhan Mantri Fasal Bima Yojana

- (A) I, II, III, IV
- (B) II, I, III, IV
- (C) I, II, IV, III
- (D) I, III, II, IV

IG06

Minerals, Industries and Transport of India

45 questions · 28 fact-sheet points

Must-know facts

1. Iron ore: Odisha is the largest producer; belts are Odisha-Jharkhand (Badampahar, Gua, Noamundi), Durg-Bastar-Chandrapur (Bailadila, Dalli-Rajhara), Bellary-Chitradurga-Tumakuru (Kudremukh) and Maharashtra-Goa.
2. Bailadila (Dantewada, Chhattisgarh) haematite is exported to Japan through Visakhapatnam; Kudremukh (Karnataka) magnetite moved through New Mangalore.
3. Manganese: Madhya Pradesh is the leading producer (about 31% as per the Indian Bureau of Mines), followed by Maharashtra and Odisha; Odisha holds the largest reserves (about 34%). Belt: Balaghat (MP) and Nagpur-Bhandara (Maharashtra).
4. Copper: Madhya Pradesh (Malanjkhand, Balaghat) is the largest producer; Rajasthan (Khetri) holds over half the country's copper reserves; Singhbhum (Jharkhand) is the third belt.
5. Bauxite: Odisha leads (Panchpatmali in Koraput, Kalahandi); other areas are Lohardaga (Jharkhand), Amarkantak plateau and Gujarat. NALCO has its refinery at Damanjodi and smelter at Angul.
6. Lead-zinc: almost the entire output is from Rajasthan — Rampura Agucha (Bhilwara), Zawar (Udaipur) and Rajpura-Dariba, worked by Hindustan Zinc.
7. Gold: Karnataka — Hutti and Kolar (KGF closed in 2001) — plus Ramagiri in Andhra Pradesh. Chromite: Odisha's Sukinda valley yields nearly all of India's output. Uranium: Jaduguda (Jharkhand) and Tummalapalle (Andhra Pradesh).
8. Limestone: Rajasthan is the largest producer (about 22%) while Karnataka holds the largest reserves; rock phosphate — Jhamarkotra (Udaipur); gypsum — Nagaur and Bikaner; mica — Koderma-Giridih (Jharkhand), Nellore (Andhra Pradesh), Bhilwara.
9. Coal: about 96% of reserves are Gondwana coal; Jharia is the largest coalfield and the chief source of prime coking coal; Raniganj (1774) is the oldest; others are Bokaro, Karanpura, Korba, Singrauli and Talcher. Jharkhand has the largest reserves; Odisha has led production in recent years.
10. Lignite (brown coal): Neyveli in Tamil Nadu is the largest field; others are Kapurdi-Jalipa and Barsingsar (Rajasthan) and Panandhro (Gujarat).
11. Petroleum: Mumbai High (found 1974, drilled by Sagar Samrat) is the largest field; Rajasthan's Barmer basin (Mangala field, 2004) makes it the leading onshore crude producer; Gujarat has Ankleshwar and Kalol; Assam has Digboi, Naharkatiya and Moran.
12. Refineries: Digboi (1901) is the oldest; Jamnagar in Gujarat is the world's largest refining complex; the HPCL Rajasthan Refinery is at Pachpadra in Balotra (erstwhile Barmer). Natural gas moves through the Hazira-Vijaipur-Jagdishpur (HVJ) pipeline of GAIL.
13. Nuclear power: Tarapur (Maharashtra, 1969) is the oldest; Rawatbhata (Rajasthan) was built with Canadian help; Kalpakkam and Kudankulam (largest, Russian-aided) are in Tamil Nadu; Kakrapar (Gujarat) has India's first indigenous 700 MWe PHWR; also Narora (UP) and Kaiga (Karnataka).
14. Steel: the Tata plant at Jamshedpur (founded 1907, production from 1912) on the Subarnarekha was India's first modern steel plant; SAIL plants — Bhilai and Bokaro (USSR), Rourkela (West Germany), Durgapur (United Kingdom); Visakhapatnam (RINL) is the only shore-based integrated plant.
15. Aluminium: Odisha leads; NALCO (Angul/Damanjodi), BALCO (Korba) and HINDALCO (Renukoot) are the main producers.
16. Cotton textiles: the first mill was set up at Mumbai in 1854; Ahmedabad is the 'Manchester of India' and Coimbatore the 'Manchester of South India'. The first jute mill came up at Rishra on the Hooghly in 1855.

17. Sugar: Uttar Pradesh has the most mills and cane area; Maharashtra has higher recovery. Cement: the first plant was set up at Chennai in 1904; India is the world's second largest cement producer. Sindri (1951) was the first large fertiliser plant.
18. Automobiles: Chennai is the 'Detroit of India'; other hubs are Gurugram-Manesar, Pune-Chakan, Sanand and Pithampur. Bengaluru is the 'Silicon Valley of India'.
19. Eight major industrial regions: Mumbai-Pune, Hugli (oldest), Bengaluru-Tamil Nadu, Gujarat, Chotanagpur, Vishakhapatnam-Guntur, Gurugram-Delhi-Meerut and Kollam-Thiruvananthapuram.
20. Dedicated Freight Corridors: Western DFC Dadri (UP) to JNPT Mumbai, about 1,506 km, with Japanese assistance; Eastern DFC Sahnewal (Ludhiana) to Sonnagar (Bihar), about 1,337 km, with World Bank assistance. The Delhi-Mumbai Industrial Corridor runs along the Western DFC and about 39% of its alignment lies in Rajasthan.
21. Roads: India's road network is the second largest in the world; national highways are about 1.46 lakh km (2024-25), roughly 2% of road length but carrying about 40% of traffic. NHAI was constituted under an Act of 1988 and became operational in 1995; Bharatmala Pariyojana (2017) Phase-I covers 34,800 km.
22. Golden Quadrilateral: 5,846 km linking Delhi, Kolkata, Chennai and Mumbai under NHDP Phase-I, completed in 2012; Delhi-Kolkata is NH-19, Kolkata-Chennai NH-16, Chennai-Mumbai and Mumbai-Delhi NH-48.
23. North-South Corridor: Srinagar to Kanniyakumari (with a Salem-Kochi spur), largely NH-44; East-West Corridor: Silchar to Porbandar, largely NH-27. The two corridors cross at Jhansi in Uttar Pradesh; both are NHDP Phase-II.
24. NH-44 (4,113 km) is the longest national highway and NH-27 (about 3,507 km) the second longest; NH-66 runs along the west coast. Under the 2010 renumbering, north-south highways get even numbers increasing from east to west and east-west highways odd numbers increasing from north to south.
25. Railways: the first passenger train ran from Bori Bunder to Thane (34 km) on 16 April 1853; Indian Railways now has 19 zones, the newest being South Coast Railway with headquarters at Visakhapatnam, functional from 1 June 2026; Kolkata Metro is counted as a zone.
26. Ports: there are 12 major ports, with VadHAVAN (Palghar, Maharashtra) approved in 2024 as a new one. Jawaharlal Nehru Port (Nhava Sheva) is the largest container-handling major port, Kolkata (Syama Prasad Mookerjee) the only major riverine port, Kamarajar (Ennore) the first corporatised major port and Kandla is now Deendayal Port.
27. Inland waterways: NW-1 Ganga, Haldia to Prayagraj (1,620 km); NW-2 Brahmaputra, Dhubri to Sadiya (891 km); NW-3 West Coast Canal, Kollam to Kottapuram (205 km). The National Waterways Act, 2016 declared 111 national waterways; the Inland Waterways Authority of India (1986) has its headquarters at Noida.
28. Industrial corridors and programmes: DMIC (Japan-aided), Chennai-Bengaluru, Amritsar-Kolkata and Vizag-Chennai corridors; Sagarmala (2015) for port-led development; Make in India (2014) and PM Gati Shakti (2021) for multimodal connectivity.

1. Which state is the largest producer of iron ore in India?

- (A) Jharkhand
(C) Odisha

- (B) Chhattisgarh
(D) Karnataka

2. The Bailadila iron ore mines are located in which state?

- (A) Chhattisgarh
(C) Jharkhand

- (B) Odisha
(D) Karnataka

3. As per the Indian Bureau of Mines, which state is the leading producer of manganese ore in India?
(A) Odisha (B) Madhya Pradesh
(C) Maharashtra (D) Karnataka
4. The Balaghat and Nagpur-Bhandara manganese belt lies in which two states? **HARD**
(A) Madhya Pradesh and Maharashtra (B) Odisha and Jharkhand
(C) Karnataka and Goa (D) Andhra Pradesh and Telangana
5. The Malanjkhand copper project is located in which state?
(A) Rajasthan (B) Jharkhand
(C) Madhya Pradesh (D) Andhra Pradesh
6. Panchpatmali, India's most important bauxite mining area, is located in which state?
(A) Jharkhand (B) Gujarat
(C) Chhattisgarh (D) Odisha
7. Rampura Agucha, India's largest lead-zinc mine, is located in which state?
(A) Rajasthan (B) Gujarat
(C) Andhra Pradesh (D) Telangana
8. Under the national highway numbering system notified in 2010, highways running north-south are given **HARD**
(A) odd numbers increasing from north to south
(B) even numbers increasing from east to west
(C) odd numbers increasing from east to west
(D) even numbers increasing from north to south
9. Which is the oldest coalfield of India, where mining began in 1774?
(A) Jharia (B) Bokaro
(C) Korba (D) Raniganj
10. The Jharia coalfield is especially important because it produces
(A) lignite (B) anthracite
(C) peat (D) prime coking coal
11. Which one of the following lignite fields is located in Rajasthan? **RAS PRE 2024**
(A) Neyveli (B) Panandhro
(C) Kapurdi-Jalipa (D) Talcher
12. The Mangala oil field, India's biggest onshore oil discovery in recent decades, lies in which basin? **RAS PRE 2013, 2018**
(A) Cambay basin (B) Barmer basin
(C) Krishna-Godavari basin (D) Assam-Arakan basin
13. Which is the largest oil field of India?
(A) Ankleshwar (B) Digboi
(C) Mumbai High (D) Nagayalanka
14. India's oldest oil refinery, set up in 1901, is located at
(A) Digboi (B) Trombay
(C) Barauni (D) Koyali

- 15.** India's first nuclear power station, commissioned in 1969, is located at
(A) Rawatbhata (B) Kalpakkam
(C) Narora (D) Tarapur
- 16.** India's first indigenously designed 700 MWe pressurised heavy water reactor was commissioned at **HARD**
(A) Kakrapar (B) Kaiga
(C) Kudankulam (D) Narora
- 17.** The Rourkela steel plant was set up with the collaboration of which country?
(A) USSR (B) United Kingdom
(C) Japan (D) West Germany
- 18.** The first cotton textile mill of India was set up in 1854 at
(A) Ahmedabad (B) Mumbai
(C) Kanpur (D) Surat
- 19.** Which city is known as the 'Detroit of India' because of its automobile industry?
(A) Pune (B) Gurugram
(C) Sanand (D) Chennai
- 20.** The Golden Quadrilateral highway network connects which four cities? **RAS PRE 2013**
(A) Delhi, Kolkata, Chennai and Mumbai (B) Delhi, Lucknow, Kolkata and Guwahati
(C) Mumbai, Bengaluru, Hyderabad and Chennai (D) Delhi, Jaipur, Ahmedabad and Mumbai
- 21.** The North-South and East-West corridors of the national highway network intersect at **RAS PRE 2016, 2021**
(A) Nagpur (B) Jhansi
(C) Bhopal (D) Kanpur
- 22.** The East-West Corridor of the National Highways Development Project connects **RAS PRE 2024**
(A) Silchar and Porbandar (B) Guwahati and Ahmedabad
(C) Kolkata and Mumbai (D) Dibrugarh and Dwarka
- 23.** What is the approximate total length of the Golden Quadrilateral? **RAS PRE 2023**
(A) 3,846 km (B) 4,846 km
(C) 5,846 km (D) 7,846 km
- 24.** Which is the longest national highway of India?
(A) NH-27 (B) NH-44
(C) NH-48 (D) NH-16
- 25.** Consider the following statements about the highway corridors of India:
I. The North-South Corridor connects Srinagar with Kanniyakumari.
II. The East-West Corridor connects Silchar with Porbandar.
III. Both corridors were taken up under Phase-II of the National Highways Development Project.
Which of the statements given above are correct? **RAS PRE 2021**
(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

26. Consider the following statements about national highways in India:

- I. They form about 2 per cent of the total road length but carry about 40 per cent of road traffic.
- II. The national highway network of the country exceeds one lakh kilometres.
- III. The National Highways Authority of India was constituted under an Act of 1988.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

27. Consider the following statements about Indian Railways:

- I. India's first passenger train ran between Bori Bunder and Thane in 1853.
- II. Kolkata Metro is counted as a zone of Indian Railways.
- III. The headquarters of the newly created South Coast Railway zone is at Vijayawada.

Which of the statements given above are correct? **HARD**

- (A) II and III only
- (B) I and III only
- (C) I and II only
- (D) I, II and III

28. Consider the following statements about the ports of India:

- I. Jawaharlal Nehru Port is the largest container-handling major port of the country.
- II. Kolkata (Syama Prasad Mookerjee) Port is India's only major riverine port.
- III. Vadhavan in Maharashtra has been approved as a new major port.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

29. Consider the following statements about coal in India:

- I. Gondwana coal accounts for the bulk of India's coal reserves.
- II. Rajasthan has the largest coal reserves in the country.
- III. Tertiary coal is found in the north-eastern states.

Which of the statements given above are correct? **HARD**

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

30. Consider the following statements about the national waterways of India:

- I. National Waterway-1 lies on the Ganga between Haldia and Prayagraj.
- II. National Waterway-2 lies on the Brahmaputra between Dhubri and Sadiya.
- III. The Inland Waterways Authority of India has its headquarters at Kolkata.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

31. Consider the following statements about the industrial regions of India:

- I. The Hugli region is the oldest industrial region of India.
- II. The Chotanagpur region is noted for heavy metallurgical industries.
- III. The Kollam-Thiruvananthapuram region is based mainly on iron and steel.

Which of the statements given above are correct? **HARD**

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

32. Consider the following statements about the Dedicated Freight Corridors:

- I. The Western Corridor runs from Dadri to Jawaharlal Nehru Port.
- II. The Eastern Corridor runs from Sahnewal (Ludhiana) to Sonnagar.
- III. The Delhi-Mumbai Industrial Corridor is being developed along the Western Corridor.

Which of the statements given above are correct?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

33. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Segment of the Golden Quadrilateral) List-II (National Highway)

- A. Delhi-Kolkata 1. NH-19
- B. Kolkata-Chennai 2. NH-16
- C. Chennai-Mumbai 3. NH-48
- D. Srinagar-Kanniyakumari 4. NH-44 **RAS PRE 2016**

- (A) A-1, B-2, C-3, D-4
- (B) A-2, B-1, C-3, D-4
- (C) A-1, B-2, C-4, D-3
- (D) A-4, B-2, C-3, D-1

34. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Mineral) List-II (Mining area)

- A. Rock phosphate 1. Jhamarkotra
- B. Chromite 2. Sukinda
- C. Diamond 3. Panna
- D. Uranium 4. Jaduguda

- (A) A-2, B-1, C-3, D-4
- (B) A-1, B-2, C-4, D-3
- (C) A-1, B-2, C-3, D-4
- (D) A-3, B-2, C-1, D-4

35. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Steel plant) List-II (State)

- A. Bhilai 1. Chhattisgarh
 B. Rourkela 2. Odisha
 C. Durgapur 3. West Bengal
 D. Bokaro 4. Jharkhand

- (A) A-2, B-1, C-3, D-4
 (B) A-1, B-2, C-4, D-3
 (C) A-4, B-2, C-3, D-1
 (D) A-1, B-2, C-3, D-4

36. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Port) List-II (State)

- A. Paradip 1. Odisha
 B. Mormugao 2. Goa
 C. Deendayal 3. Gujarat
 D. V. O. Chidambaranar 4. Tamil Nadu

- (A) A-2, B-1, C-3, D-4
 (B) A-1, B-2, C-4, D-3
 (C) A-1, B-2, C-3, D-4
 (D) A-3, B-2, C-1, D-4

37. Match List-I with List-II and select the correct answer using the codes given below:

List-I (Industrial region) List-II (Characteristic industry)

- A. Hugli 1. Jute
 B. Chotanagpur 2. Iron and steel
 C. Gujarat 3. Cotton textiles and petrochemicals
 D. Vishakhapatnam-Guntur 4. Shipbuilding HARD

- (A) A-1, B-2, C-3, D-4
 (B) A-2, B-1, C-3, D-4
 (C) A-1, B-2, C-4, D-3
 (D) A-4, B-2, C-3, D-1

38. Which one of the following mineral-state pairs is NOT correctly matched?

- (A) Bauxite — Punjab (B) Mica — Jharkhand
 (C) Gold — Karnataka (D) Chromite — Odisha

39. Which one of the following is NOT a nuclear power station of India?

- (A) Kaiga (B) Neyveli
 (C) Narora (D) Kakrapar

40. Which one of the following national highway-route pairs is NOT correctly matched?

- (A) NH-44 — Srinagar to Kanniyakumari (B) NH-27 — Porbandar to Silchar
 (C) NH-48 — Delhi to Kolkata (D) NH-16 — Kolkata to Chennai

41. Which one of the following is NOT a major port of India?

- (A) Paradip (B) New Mangalore
 (C) Kamarajar (D) Mundra

42. Which one of the following industrial region-state pairs is NOT correctly matched?

- (A) Chotanagpur — Chhattisgarh
- (B) Mumbai-Pune — Maharashtra
- (C) Hugli — West Bengal
- (D) Kollam-Thiruvananthapuram — Kerala

43. Which one of the following coalfield-state pairs is NOT correctly matched? **HARD**

- (A) Talcher — Odisha
- (B) Bokaro — West Bengal
- (C) Korba — Chhattisgarh
- (D) Singrauli — Madhya Pradesh

44. Arrange the following steel plants in the chronological order in which they started production:

I. Jamshedpur

II. Bhilai

III. Bokaro

IV. Visakhapatnam **HARD**

- (A) II, I, III, IV
- (B) I, III, II, IV
- (C) I, II, III, IV
- (D) I, II, IV, III

45. Assertion (A): Odisha is the leading producer of manganese ore in India.

Reason (R): Odisha holds the largest share of the country's manganese ore reserves.

Select the correct answer using the codes given below: **HARD**

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is not the correct explanation of A
- (C) A is true but R is false
- (D) A is false but R is true

Answer key & explanations

IG01 – PHYSIOGRAPHIC DIVISIONS OF INDIA

IG01.1 D	IG01.11 C	IG01.21 A	IG01.31 B
IG01.2 C	IG01.12 B	IG01.22 C	IG01.32 A
IG01.3 A	IG01.13 B	IG01.23 A	IG01.33 C
IG01.4 D	IG01.14 D	IG01.24 C	IG01.34 D
IG01.5 B	IG01.15 A	IG01.25 B	IG01.35 B
IG01.6 C	IG01.16 A	IG01.26 C	IG01.36 C
IG01.7 B	IG01.17 D	IG01.27 A	IG01.37 D
IG01.8 A	IG01.18 A	IG01.28 D	IG01.38 A
IG01.9 C	IG01.19 D	IG01.29 B	IG01.39 B
IG01.10 D	IG01.20 C	IG01.30 D	IG01.40 B

IG02 – CLIMATE OF INDIA

IG02.1 C	IG02.10 B	IG02.19 C	IG02.28 A
IG02.2 D	IG02.11 D	IG02.20 D	IG02.29 D
IG02.3 B	IG02.12 A	IG02.21 A	IG02.30 B
IG02.4 C	IG02.13 D	IG02.22 B	IG02.31 D
IG02.5 C	IG02.14 A	IG02.23 C	IG02.32 C
IG02.6 A	IG02.15 B	IG02.24 D	IG02.33 B
IG02.7 D	IG02.16 B	IG02.25 A	IG02.34 C
IG02.8 A	IG02.17 C	IG02.26 B	IG02.35 A
IG02.9 B	IG02.18 D	IG02.27 C	

IG03 – RIVERS, LAKES AND DRAINAGE OF INDIA

IG03.1 B	IG03.11 A	IG03.21 A	IG03.31 B
IG03.2 C	IG03.12 A	IG03.22 B	IG03.32 D
IG03.3 C	IG03.13 D	IG03.23 C	IG03.33 C
IG03.4 C	IG03.14 D	IG03.24 A	IG03.34 B
IG03.5 D	IG03.15 A	IG03.25 C	IG03.35 A
IG03.6 A	IG03.16 B	IG03.26 B	IG03.36 D
IG03.7 A	IG03.17 C	IG03.27 D	IG03.37 C
IG03.8 D	IG03.18 C	IG03.28 A	IG03.38 B
IG03.9 D	IG03.19 D	IG03.29 B	IG03.39 A
IG03.10 A	IG03.20 B	IG03.30 C	IG03.40 D

IG04 – IRRIGATION AND MULTIPURPOSE PROJECTS OF INDIA

IG04.1 C	IG04.9 C	IG04.17 D	IG04.25 B
IG04.2 D	IG04.10 C	IG04.18 B	IG04.26 D
IG04.3 C	IG04.11 B	IG04.19 A	IG04.27 B
IG04.4 B	IG04.12 B	IG04.20 B	IG04.28 D
IG04.5 C	IG04.13 D	IG04.21 C	IG04.29 A
IG04.6 A	IG04.14 D	IG04.22 D	IG04.30 A
IG04.7 B	IG04.15 A	IG04.23 A	
IG04.8 A	IG04.16 A	IG04.24 C	

IG05 – AGRICULTURE AND CROPS OF INDIA

IG05.1 C	IG05.2 D	IG05.3 C	IG05.4 D
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IG05.5 A	IG05.14 C	IG05.23 B	IG05.32 C
IG05.6 A	IG05.15 C	IG05.24 D	IG05.33 D
IG05.7 D	IG05.16 B	IG05.25 A	IG05.34 B
IG05.8 A	IG05.17 B	IG05.26 B	IG05.35 C
IG05.9 B	IG05.18 C	IG05.27 B	IG05.36 D
IG05.10 C	IG05.19 B	IG05.28 D	IG05.37 C
IG05.11 B	IG05.20 A	IG05.29 A	IG05.38 B
IG05.12 C	IG05.21 B	IG05.30 C	IG05.39 D
IG05.13 B	IG05.22 C	IG05.31 A	IG05.40 A

IG06 – MINERALS, INDUSTRIES AND TRANSPORT OF INDIA

IG06.1 C	IG06.13 C	IG06.25 D	IG06.37 A
IG06.2 A	IG06.14 A	IG06.26 D	IG06.38 A
IG06.3 B	IG06.15 D	IG06.27 C	IG06.39 B
IG06.4 A	IG06.16 A	IG06.28 D	IG06.40 C
IG06.5 C	IG06.17 D	IG06.29 B	IG06.41 D
IG06.6 D	IG06.18 B	IG06.30 A	IG06.42 A
IG06.7 A	IG06.19 D	IG06.31 A	IG06.43 B
IG06.8 B	IG06.20 A	IG06.32 D	IG06.44 C
IG06.9 D	IG06.21 B	IG06.33 A	IG06.45 D
IG06.10 D	IG06.22 A	IG06.34 C	
IG06.11 C	IG06.23 C	IG06.35 D	
IG06.12 B	IG06.24 B	IG06.36 C	

IG01 – EXPLANATIONS

1. (D) Kangchenjunga — Kangchenjunga (8,586 m) is the highest peak of India and the third highest in the world after Mount Everest and K2.

2. (C) Nanda Devi — Nanda Devi (7,816 m) in Chamoli district, Uttarakhand, is the highest peak wholly within India. Kangchenjunga is higher but straddles the India-Nepal border.

3. (A) Kerala — Anamudi (2,695 m) stands at the junction of the Anaimalai, Cardamom and Palani hills in Idukki-Ernakulam districts of Kerala. It is the highest point in India south of the Himalaya.

4. (D) Andhra Pradesh — Arma Konda (1,680 m) in Alluri Sitharama Raju district of Andhra Pradesh is the highest peak of the Eastern Ghats. Mahendragiri in Odisha is another prominent Eastern Ghats peak.

5. (B) Nilgiri hills — Doddabetta near Udthagamandalam (Ooty) in Tamil Nadu is the highest peak of the Nilgiris, the block where the Eastern and Western Ghats converge.

6. (C) Nilgiri hills — The two Ghats converge in the Nilgiri hills. South of the Nilgiris the Palghat gap breaks the continuity of the Western Ghats.

7. (B) Barren Island — Barren Island in the Andaman Sea is the only active volcano in Indian territory; Narcondam Island nearby is dormant.

8. (A) Little Andaman from Car Nicobar — The Ten Degree Channel divides the Andaman group from the Nicobar group, i.e. Little Andaman from Car Nicobar. The Duncan Passage lies between South Andaman and Little Andaman.

9. (C) Great Nicobar — Indira Point (6°45'N) lies at the southern tip of Great Nicobar. It was earlier called Pygmalion Point and subsided during the 2004 tsunami.

10. (D) Mirzapur — The 82°30'E meridian passes through Mirzapur in Uttar Pradesh and fixes Indian Standard Time at GMT + 5:30.

11. (C) Eight — It crosses Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram — eight states.

- 12. (B) Bangladesh** — The India-Bangladesh boundary is 4,096.7 km long, the longest; the shortest is with Afghanistan (106 km).
- 13. (B) Bhabar** — The Bhabar is an 8–16 km wide pebble belt at the foot of the Shivalik. South of it the streams re-emerge in the marshy Terai.
- 14. (D) Khadar** — Khadar is the new alluvium renewed almost every year by floods; Bhangar is the older alluvium and often contains calcareous kankar nodules.
- 15. (A) Duns** — Dehra Dun, Kotli Dun and Patli Dun are such valleys. Karewas are lacustrine deposits of the Kashmir valley and bugyals are alpine meadows.
- 16. (A) Sirohi** — Guru Shikhar (1,722 m) is on the Mount Abu plateau in Sirohi district.
- 17. (D) Basalt** — The Deccan Trap was formed by fissure eruptions of basaltic lava of Cretaceous-Eocene age; its weathering produces black regur soil.
- 18. (A) Malda gap** — The Malda gap, also called the Garo-Rajmahal gap, was filled by later Ganga-Brahmaputra alluvium and detaches the Meghalaya plateau from the Chotanagpur block.
- 19. (D) Gujarat** — Gujarat has the longest coastline of any Indian state — 2,340.62 km in the revised National Hydrographic Office assessment.
- 20. (C) 11,098.81 km** — The earlier official figure of 7,516.6 km was revised to 11,098.81 km using high-resolution mapping. 15,106.7 km is the length of India's land frontier.
- 21. (A) Srinagar and Leh** — Zoji La on NH-1 crosses the Great Himalaya and links the Kashmir valley with Kargil and Leh.
- 22. (C) Zaskar** — The trans-Himalayan sequence from north to south is Karakoram – Ladakh – Zaskar, and then the Great Himalaya. The Pir Panjal and Dhauladhar belong to the Lesser Himalaya, south of the Himadri.
- 23. (A) I and II only** — The Shivalik averages only 900–1,100 m, so statement III is wrong. Its outer, unconsolidated character makes it highly prone to landslides and erosion.
- 24. (C) Both I and II** — Both are correct. Much of the Terai in Uttarakhand and Uttar Pradesh was cleared for agriculture after Partition.
- 25. (B) I, II and III** — All three are correct. The Peninsular block is one of the oldest and most stable landmasses on earth.
- 26. (C) I and II only** — The Duncan Passage separates South Andaman from Little Andaman; it is the Ten Degree Channel that separates Little Andaman from Car Nicobar, so III is wrong.
- 27. (A) I, II and III** — All three are correct. The Nine Degree Channel separates Minicoy from the rest of the Lakshadweep group.
- 28. (D) I, II and III** — All three are correct. The submerged western coast provides natural harbours such as Mumbai and Marmagao, while the emergent eastern coast has few good ports.
- 29. (B) I, II and III** — All three are correct. The Aravalli survives today as a heavily denuded relict range, with detached Vindhyan outliers to its east.
- 30. (D) I, II and III** — All three are correct in Sydney Burrard's fourfold division; the Nepal Himalaya lies between the Kali and the Teesta.
- 31. (B) A-2, B-1, C-4, D-3** — Shipki La is in Kinnaur (HP) and admits the Satluj into India; Nathu La is in Sikkim; Se La lies on the road to Tawang in Arunachal Pradesh; Lipulekh is in Pithoragarh, Uttarakhand.
- 32. (A) A-4, B-1, C-2, D-3** — Anamudi (2,695 m) belongs to the Anaimalai group, Doddabetta (2,637 m) to the Nilgiris, Guru Shikhar (1,722 m) to the Aravalli and Dhupgarh (1,350 m) to the Satpura.
- 33. (C) A-2, B-3, C-4, D-1** — The Cardamom hills lie in Kerala, the Javadi hills of the Eastern Ghats in Tamil Nadu, the Rajmahal hills in Jharkhand and the Mukundara hills near Kota in Rajasthan.
- 34. (D) A-2, B-3, C-4, D-1** — The Ten Degree Channel divides the Andaman and Nicobar groups; the Duncan Passage lies within the Andamans; the Nine and Eight Degree Channels lie in the Lakshadweep sea.
- 35. (B) A-4, B-1, C-2, D-3** — The Punjab doabs from east to west are Bist, Bari, Rechna, Chaj and Sindh Sagar; the last lies between the Jhelum and the Indus.

- 36. (C) Anamudi — Tamil Nadu** — Anamudi is in Kerala, at the meeting point of the Anaimalai, Cardamom and Palani hills; Doddabetta is the highest peak in Tamil Nadu.
- 37. (D) Kaimur hills** — The Kaimur hills belong to the Vindhyan system of the Peninsular Plateau. The Purvanchal comprises the Patkai Bum, Naga, Manipur, Mizo, Barail and Garo-Khasi-Jaintia hills.
- 38. (A) The Khadar belt lies away from the flood plains and is not renewed by fresh silt.** — It is the Bhangar that lies above the flood level and is not renewed; the Khadar is the new alluvium of the flood plains, enriched almost every year.
- 39. (B) Bhore Ghat — Karnataka** — The Bhore Ghat is in Maharashtra and carries the Mumbai-Pune route; the Thal Ghat carries the Mumbai-Nashik route.
- 40. (B) II, IV, III, I** — The correct order is Karakoram - Ladakh - Zaskar - Great Himalaya - Lesser Himalaya - Shivalik, i.e. II, IV, III, I.

IG02 – EXPLANATIONS

- 1. (C) Mediterranean Sea** — Western disturbances are extra-tropical cyclones formed over the Mediterranean region and steered into India by the subtropical westerly jet stream.
- 2. (D) Mahawat** — Mahawat is small in amount but extremely important for the rabi wheat crop of Punjab, Haryana and adjoining Rajasthan.
- 3. (B) 1 June** — IMD's normal onset date over Kerala is 1 June; the monsoon normally covers the whole country by about 8 July.
- 4. (C) 17 September** — In 2020 IMD revised the date from 1 September to 17 September; final withdrawal from the country is complete by 15 October.
- 5. (C) Retreating (north-east) monsoon** — During the retreating monsoon the north-east winds pick up moisture while crossing the Bay of Bengal and shed it on the Coromandel coast.
- 6. (A) It lies in the rain-shadow of the Western Ghats and runs parallel to the Bay of Bengal branch** — The Arabian Sea branch has already shed its moisture on the Western Ghats, while the Bay branch moves parallel to the coast without striking it.
- 7. (D) Mawsynram** — Mawsynram in the Khasi hills of Meghalaya receives the highest average annual rainfall in India and in the world; Cherrapunji (Sohra) lies close by.
- 8. (A) A hot, dry westerly wind of the northern plains in May and June** — The Loo blows in the afternoons of May-June across the northern plains and can raise temperatures to 45-47 degrees Celsius.
- 9. (B) West Bengal** — Kal Baisakhi (the nor'wester) strikes West Bengal in April-May; the same storms are called Bardoli Chheerha in Assam.
- 10. (B) Kerala and coastal Karnataka** — These April-May showers help the early ripening of mangoes; in the coffee areas of Karnataka they are called blossom showers.
- 11. (D) Intense heating of the Tibetan plateau** — Heating of the Tibetan plateau creates a high-level anticyclone from which the tropical easterly jet flows out over peninsular India, aiding the monsoon.
- 12. (A) Somali (Findlater) jet** — The Somali or Findlater jet flows past the East African coast and turns into the Arabian Sea, feeding moisture into the monsoon current.
- 13. (D) Deficient monsoon rainfall over India** — El Nino, the warming of the eastern equatorial Pacific, weakens the Indian monsoon, while La Nina generally strengthens it.
- 14. (A) Tahiti and Darwin** — A persistently negative SOI (lower pressure at Tahiti than Darwin) indicates El Nino conditions and a likely weak Indian monsoon.
- 15. (B) The mechanism of the monsoon** — The Monsoon Experiment was carried out under the Global Weather Experiment — Winter MONEX in December 1978 and Summer MONEX in May-July 1979 over the Arabian Sea and the Bay of Bengal.
- 16. (B) BWhw** — BWhw denotes a hot desert climate with dry winters, covering Barmer, Jaisalmer and adjoining western Rajasthan; BShw is the semi-arid steppe around it.
- 17. (C) Western Rajasthan and Ladakh** — Variability exceeds 50% in west Rajasthan, Ladakh and parts of the interior Deccan; it is under 25% where rainfall is heavy, such as the west coast and the north-east.

- 18. (D) 1875** — IMD was set up in 1875 and is the national meteorological service of India, headquartered at New Delhi.
- 19. (C) New Delhi** — IMD, New Delhi, acts as the RSMC for tropical cyclones over the North Indian Ocean and issues names on behalf of thirteen WMO/ESCAP member countries.
- 20. (D) I, II and III** — All three are correct. Western disturbances also cause snowfall in the western Himalaya and occasional hailstorms in Rajasthan.
- 21. (A) I, II and III** — All three are correct. The deflection in statement II is caused by the Coriolis force in the northern hemisphere.
- 22. (B) I, II and III** — All three are correct. The retreating monsoon season also sees frequent tropical cyclones over the Bay of Bengal.
- 23. (C) I, II and III** — All three are correct. A strong positive IOD can partly offset the drying effect of an El Nino year.
- 24. (D) I, II and III** — All three are correct. Mumbai is fed by the Arabian Sea branch in the south-west monsoon, while Chennai depends largely on the north-east monsoon.
- 25. (A) I, II and III** — All three are correct. The pre-monsoon (April-June) and post-monsoon (October-December) periods are the two cyclone peaks.
- 26. (B) II and III only** — Statement I is wrong: a deficiency of 26–50% of normal rainfall is a moderate drought and more than 50% a severe drought.
- 27. (C) A-3, B-4, C-2, D-1** — The Loo is a hot summer wind of the northern plains, Kal Baisakhi a nor'wester of West Bengal, mango showers pre-monsoon rain of Kerala and coastal Karnataka, and Bardoli Chheerha the nor'wester of Assam.
- 28. (A) A-2, B-4, C-1, D-3** — Amw is the monsoon type with a short dry season on the west coast, As the monsoon type with dry summer on the Coromandel coast, BWhw the hot desert of west Rajasthan and Dfc the cold humid winter type of Arunachal Pradesh.
- 29. (D) A-3, B-1, C-4, D-2** — IMD divides the year into winter (January-February), pre-monsoon (March-May), south-west monsoon (June-September) and post-monsoon (October-December).
- 30. (B) A-3, B-1, C-2, D-4** — The northward shift of the subtropical westerly jet, the establishment of the tropical easterly jet and the strengthening of the Somali jet together mark the onset of the monsoon.
- 31. (D) Bardoli Chheerha — Punjab** — Bardoli Chheerha is the nor'wester of Assam, not of Punjab.
- 32. (C) The Arabian Sea branch causes maximum rainfall over the Coromandel coast.** — The Coromandel coast lies in the rain-shadow of the Western Ghats and gets little rain in the south-west monsoon; it is watered mainly by the retreating monsoon.
- 33. (B) They reach India along with the tropical easterly jet stream.** — Western disturbances are steered by the subtropical westerly jet stream, not the tropical easterly jet, which is a summer feature of peninsular India.
- 34. (C) Aw — Arunachal Pradesh** — Arunachal Pradesh falls under Dfc, the cold humid winter type with a short summer. Aw is the tropical savanna type of most of the peninsula.
- 35. (A) Both A and R are true and R is the correct explanation of A** — The retreating monsoon blows offshore over most of India, but over the Bay of Bengal it gathers moisture and sheds it on the Coromandel coast.

IG03 – EXPLANATIONS

- 1. (B) Ghaghara** — The Rapti rises in Nepal and joins the Ghaghara in eastern Uttar Pradesh. The Ghaghara itself is the Karnali in Nepal and is a left-bank tributary of the Ganga.
- 2. (C) Brahmaputra** — The Teesta drains Sikkim and northern West Bengal and joins the Brahmaputra in Bangladesh. It once flowed into the Ganga but shifted course after the flood of 1787.
- 3. (C) Narmada** — The Hiran joins the Narmada in Madhya Pradesh. Other Narmada tributaries are the Burhner, Banjar, Tawa, Sher, Barna, Kolar and Orsang.
- 4. (C) Godavari** — This Purna is a right-bank tributary of the Godavari in Marathwada. A different river of the same name in Vidarbha joins the Tapi, which is why the district is specified.

- 5. (D) Devprayag** — The two headstreams meet at Devprayag in Tehri Garhwal, below which the river is called the Ganga. It debouches on to the plains at Haridwar.
- 6. (A) Tandi** — The two streams meet at Tandi in Lahaul, Himachal Pradesh. The Chenab is the largest tributary of the Indus in India by volume.
- 7. (A) Verinag** — The Jhelum rises at Verinag in Anantnag district, flows through the Kashmir valley and Wular lake, and enters Pakistan through the Baramulla gorge.
- 8. (D) Karachi** — It was signed at Karachi on 19 September 1960 by Jawaharlal Nehru and Ayub Khan, with the World Bank as a signatory.
- 9. (D) Ravi, Beas and Satluj** — The three eastern rivers — Ravi, Beas and Satluj — were allotted to India and the three western rivers — Indus, Jhelum and Chenab — to Pakistan.
- 10. (A) Godavari** — The Godavari (about 1,465 km) is the longest peninsular river and is called the Dakshin Ganga; its basin is also the largest in peninsular India.
- 11. (A) Brahmagiri hills** — Talakaveri lies in the Brahmagiri range of Kodagu district, Karnataka. The Kaveri flows through Karnataka and Tamil Nadu into the Bay of Bengal.
- 12. (A) Narmada and Tapi** — The Narmada and the Tapi flow west through fault troughs and discharge into the Gulf of Khambhat as estuaries, carrying too little silt in suspension to build deltas.
- 13. (D) Mahi** — The Mahi rises in the Vindhyan hills of Madhya Pradesh, loops north into Rajasthan and turns south-west into Gujarat, cutting the Tropic of Cancer twice.
- 14. (D) Wular** — Wular lake in Bandipora district of Jammu and Kashmir, fed by the Jhelum, is India's largest freshwater lake and a Ramsar site.
- 15. (A) Vembanad** — Vembanad in Kerala stretches about 96 km and is the longest lake in India; the Nehru Trophy boat race is held on it and the Vembanad-Kol wetland is a Ramsar site.
- 16. (B) Loktak** — Keibul Lamjao lies on the floating phumdi mats of Loktak lake in Manipur and protects the sangai or brow-antlered deer.
- 17. (C) Uttarakhand** — Roopkund lies at about 5,029 m in Chamoli district of Uttarakhand, below the Trisul massif, and is famous for the human skeletal remains found in and around it.
- 18. (C) Odisha** — Chilika lies along the Odisha coast, holds the Nalabana Bird Sanctuary and shelters Irrawaddy dolphins.
- 19. (D) Brahmaputra** — Majuli in Assam lies in the Brahmaputra and is also a centre of the Vaishnavite satras founded in the tradition of Srimanta Sankardeva.
- 20. (B) Chilika and Keoladeo Ghana** — Chilika in Odisha and Keoladeo Ghana in Bharatpur, Rajasthan, were designated in 1981. Sambhar, Rajasthan's other early site, was designated in 1990.
- 21. (A) Ken and the Betwa** — The Ken-Betwa Link Project in Madhya Pradesh and Uttar Pradesh, centred on the Daudhan dam, is the first project under the National Perspective Plan of 1980.
- 22. (B) I, II and III** — All three are correct. The Ganga-Brahmaputra delta, the largest in the world, is the classic example of statement III.
- 23. (C) I, II and III** — All three are correct. India placed the treaty in abeyance in April 2025 following the Pahalgam terror attack.
- 24. (A) I and II only** — Statement III is wrong: in Bangladesh the Brahmaputra is called the Jamuna; the Padma is the name of the Ganga there, and the combined stream below their junction is the Meghna.
- 25. (C) I, II and III** — All three are correct. The Marble Rocks gorge at Bhedaghat near Jabalpur lies immediately below the Dhuandhar falls.
- 26. (B) I, II and III** — All three are correct. Chilika is separated from the Bay of Bengal by a sandbar and supports a resident population of Irrawaddy dolphins.
- 27. (D) I, II and III** — All three are correct. Sriharikota island, which carries India's spaceport, separates Pulicat lagoon from the Bay of Bengal.
- 28. (A) I, II and III** — All three are correct. The Daudhan dam on the Ken lies close to the Panna Tiger Reserve, which is why the project has been environmentally contested.

- 29. (B) A-3, B-1, C-4, D-2** — The Rapti joins the Ghaghara, the Teesta the Brahmaputra, the Hiran the Narmada and the Marathwada Purna the Godavari.
- 30. (C) A-2, B-3, C-4, D-1** — The Godavari rises at Trimbakeshwar, the Krishna near Mahabaleshwar, the Mahanadi near Sihawa in Chhattisgarh and the Tapi at Multai in Madhya Pradesh.
- 31. (B) A-2, B-1, C-4, D-3** — Kolleru lies between the Krishna and Godavari deltas in Andhra Pradesh, Loktak in Manipur, Lonar in Buldhana district of Maharashtra and Roopkund in Chamoli district of Uttarakhand.
- 32. (D) A-2, B-4, C-1, D-3** — Jog falls lie on the Sharavathi in Karnataka, Dudhsagar on the Mandovi in Goa, Chitrakote on the Indravati in Chhattisgarh and Hundru on the Subarnarekha in Jharkhand.
- 33. (C) A-3, B-4, C-1, D-2** — The Panch Prayag in order are Vishnuprayag (Dhauliganga), Nandprayag (Nandakini), Karnaprayag (Pindar), Rudraprayag (Mandakini) and Devprayag (Bhagirathi).
- 34. (B) Krishna — Sabari** — The Sabari is a tributary of the Godavari, not of the Krishna. Major Krishna tributaries are the Bhima, Koyna, Ghataprabha, Malaprabha, Tungabhadra and Musi.
- 35. (A) Kanwar (Kabartal) — West Bengal** — Kanwar or Kabartal lake is in Begusarai district of Bihar and is one of the largest freshwater oxbow lakes in the region.
- 36. (D) Son** — The Son is a right-bank tributary rising at Amarkantak. The Ghaghara, Gandak, Kosi, Ramganga, Gomti and Mahananda join the Ganga from the left.
- 37. (C) The Narmada and the Tapi build large deltas at their mouths.** — The Narmada and the Tapi flow through rift valleys and form estuaries, not deltas.
- 38. (B) Chitrakote — Odisha** — Chitrakote falls lie on the Indravati river in the Bastar region of Chhattisgarh and are the widest falls in India.
- 39. (A) II, I, IV, III** — Ganga about 2,525 km, Godavari about 1,465 km, Krishna about 1,400 km and Narmada about 1,312 km.
- 40. (D) III, I, IV, II** — Going downstream the order is Vishnuprayag, Nandprayag, Karnaprayag, Rudraprayag and Devprayag.

IG04 — EXPLANATIONS

- 1. (C) Sutlej** — Bhakra Dam (226 m) stands in Bilaspur district of Himachal Pradesh on the Sutlej; its reservoir is Gobind Sagar and the Nangal dam lies downstream.
- 2. (D) Bhilangana** — The 260.5 m Tehri Dam in Uttarakhand is a rock-fill dam at the Bhagirathi-Bhilangana confluence.
- 3. (C) Odisha** — Hirakud (Sambalpur, Odisha) was commissioned in 1957; with its dykes it stretches about 25.8 km.
- 4. (B) Damodar Valley Corporation** — The Damodar Valley Corporation was set up in 1948 by an Act of Parliament for Jharkhand and West Bengal, on the TVA model.
- 5. (C) Krishna** — Nagarjuna Sagar lies on the Krishna at Nandikonda on the Telangana-Andhra Pradesh border; its canals are named Jawahar and Lal Bahadur.
- 6. (A) Rihand** — The Rihand Dam in Sonbhadra district of Uttar Pradesh impounds Govind Ballabh Pant Sagar on the Rihand, a tributary of the Son.
- 7. (B) divert water into the Bhagirathi-Hooghly to keep Kolkata port navigable** — Commissioned in 1975 in West Bengal, Farakka diverts Ganga water through a feeder canal into the Bhagirathi-Hooghly to flush silt from Kolkata port.
- 8. (A) Harike** — It off-takes from the Harike barrage just below the Sutlej-Beas confluence; the Rajasthan feeder (204 km) is followed by the main canal (445 km). It was earlier called the Rajasthan Canal.
- 9. (C) Wells and tube wells** — Wells and tube wells together irrigate over 60% of the net irrigated area, tube wells alone about 45%; canals account for roughly a quarter and tanks only about 3%.
- 10. (C) Rice** — Rice is the most heavily irrigated crop in India, followed by wheat; together they take the bulk of irrigation water.
- 11. (B) Ministry of Jal Shakti** — Atal Jal is a Rs 6,000 crore central sector scheme of the Ministry of Jal Shakti, half of it funded by a World Bank loan, for community-led groundwater management.

- 12. (B) Per Drop More Crop** — 'Per Drop More Crop' covers micro-irrigation; from 2022-23 it has been implemented under the Rashtriya Krishi Vikas Yojana rather than PMKSY.
- 13. (D) 1974-75** — Launched in 1974-75 to bridge the gap between irrigation potential created and utilised, it now continues as Command Area Development and Water Management under PMKSY.
- 14. (D) Rajasthan** — The Sardar Sarovar dam on the Narmada in Gujarat supplies water to four states, Rajasthan receiving water through the Narmada main canal in Jalore and Barmer.
- 15. (A) Karnataka and Andhra Pradesh** — The Tungabhadra dam at Hospet is shared by Karnataka and Andhra Pradesh; the Tungabhadra is a tributary of the Krishna.
- 16. (A) Madhya Pradesh and Uttar Pradesh** — Water surplus in the Ken is to be transferred to the Betwa through the Daudhan dam in Panna district; the National Water Development Agency (1982) prepares such link proposals.
- 17. (D) I, II and III** — All three are correct. The DVC also operates Tilaiya and Konar dams and the Durgapur barrage, and was India's first multipurpose river valley project.
- 18. (B) I and III only** — Statement II is wrong — Atal Jal covers only seven water-stressed states: Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh.
- 19. (A) I and II only** — Statement III is wrong — the main canal runs south-west up to Mohangarh in Jaisalmer; Sriganganagar lies on the earlier Gang Canal system.
- 20. (B) I and III only** — The stage of extraction was 60.47%, and 4,951 of 6,746 units (about 73%) were 'safe', so statement II is wrong; 751 units were over-exploited.
- 21. (C) II and III only** — Tanks irrigate only about 3% of the net irrigated area and matter mainly in the peninsular plateau, so statement I is wrong. Nearly 98% of Punjab's net sown area is irrigated.
- 22. (D) I, II and III** — All three are correct; the 2021-26 phase of PMKSY has an outlay of Rs 93,068.56 crore covering AIBP, Har Khet Ko Pani and Watershed Development.
- 23. (A) A-2, B-1, C-4, D-3** — Hirakud is on the Mahanadi, Nagarjuna Sagar on the Krishna, Bhakra on the Sutlej and Gandhi Sagar on the Chambal in Madhya Pradesh.
- 24. (C) A-1, B-2, C-3, D-4** — Ukai is on the Tapi in Gujarat, Koyna in Maharashtra, Idukki an arch dam on the Periyar in Kerala, and Mettur (Stanley reservoir) on the Kaveri in Tamil Nadu.
- 25. (B) A-1, B-2, C-3, D-4** — Khadins are run-off farming embankments of Jaisalmer, zabo combines forestry and water harvesting in Nagaland, ahars are the flood-water system of south Bihar and kuhls are hill channels of Himachal.
- 26. (D) A-1, B-2, C-3, D-4** — The Kosi project is an Indo-Nepal venture for Bihar, the Gandak project additionally serves Uttar Pradesh, the Chambal valley project is shared by Madhya Pradesh and Rajasthan, and Parambikulam-Aliyar by Tamil Nadu and Kerala.
- 27. (B) Bhakra — Beas** — The Bhakra dam is on the Sutlej, not the Beas; the Beas is dammed at Pong (Maharana Pratap Sagar) and Pandoh.
- 28. (D) Pradhan Mantri Fasal Bima Yojana** — PMFBY is the crop insurance scheme and is not part of PMKSY, whose 2021-26 components are AIBP, Har Khet Ko Pani and Watershed Development.
- 29. (A) Punjab** — Despite severe groundwater depletion, Punjab is not among the seven states covered — Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh.
- 30. (A) I, II, III, IV** — DVC was created in 1948, Hirakud was completed in 1957, the Bhakra dam was dedicated to the nation in 1963 and the Farakka barrage was commissioned in 1975.

IG05 – EXPLANATIONS

- 1. (C) Mustard** — Mustard is sown in October-December and harvested in spring. Bajra, cotton and jute are kharif crops.
- 2. (D) Jute** — West Bengal produces roughly four-fifths of India's jute, grown in the Ganga-Brahmaputra delta; Bihar and Assam follow.
- 3. (C) Assam** — Assam alone produces about half of India's tea; the Brahmaputra and Barak valleys are the main belts, with West Bengal (Darjeeling, Dooars) second.

- 4. (D) Coffee** — Karnataka produces about 70% of India's coffee (Chikkamagaluru, Kodagu, Hassan), followed by Kerala and Tamil Nadu. Coffee cultivation in India began in the Baba Budan hills.
- 5. (A) Kerala** — Kerala produces the overwhelming share of India's natural rubber; Tripura is the second largest producer.
- 6. (A) Sugarcane** — Uttar Pradesh leads in both area and production of sugarcane, and is also the largest producer of wheat; sugarcane receives a Fair and Remunerative Price rather than an MSP.
- 7. (D) Rajasthan** — Rajasthan produces close to half of India's rapeseed-mustard and is also the leading producer of bajra and barley.
- 8. (A) Soybean** — Madhya Pradesh, the 'soya state', leads in soybean; Gujarat leads in groundnut and Rajasthan in rapeseed-mustard.
- 9. (B) Gujarat** — Gujarat leads in cotton production, while Maharashtra has the largest area under the crop; cotton is a kharif crop of the black soil region.
- 10. (C) Madhya Pradesh** — Madhya Pradesh leads in total pulses and in gram; Maharashtra leads in tur (arhar). India is the world's largest producer and consumer of pulses.
- 11. (B) Rajasthan** — Rajasthan leads in bajra, a hardy kharif millet of light sandy soils; jowar is led by Maharashtra and Karnataka.
- 12. (C) Alluvial soil** — Alluvial soils cover the northern plains, the coastal plains and the deltas and are the most extensive and most fertile soil group of India.
- 13. (B) basaltic lava of the Deccan trap** — Regur develops on the Deccan trap basalt of Maharashtra, Madhya Pradesh, Gujarat and adjoining areas; it is clayey, moisture-retentive and cracks in summer.
- 14. (C) Intense leaching under high rainfall and high temperature** — Alternating wet and dry spells leach away silica, leaving iron and aluminium oxides. Laterite is poor in fertility but supports tea, coffee, cashew and rubber, and is cut into building bricks.
- 15. (C) M. S. Swaminathan** — M. S. Swaminathan led the introduction of high-yielding dwarf wheat varieties from 1966-67; Norman Borlaug bred those Mexican varieties and Verghese Kurien led the White Revolution.
- 16. (B) Milk** — Operation Flood, run by the National Dairy Development Board under Verghese Kurien on the Anand cooperative pattern, made India the world's largest milk producer.
- 17. (B) 15** — The Planning Commission's 15 agro-climatic zones are used for agricultural planning; ICAR's NARP recognises 127 zones and NBSS&LUP 20 agro-ecological regions.
- 18. (C) the Commission for Agricultural Costs and Prices** — CACP (1965) recommends MSPs, which are announced by the Cabinet Committee on Economic Affairs for 22 crops; the FCI is the procurement agency.
- 19. (B) 2016** — e-NAM was launched on 14 April 2016 and is implemented by the Small Farmers' Agribusiness Consortium, integrating APMC mandis into a single online market.
- 20. (A) 1.5 per cent of the sum insured** — Farmer premium is capped at 2% for kharif, 1.5% for rabi food and oilseed crops and 5% for commercial and horticultural crops; the balance is shared by the Centre and the states.
- 21. (B) Rs 6,000** — PM-KISAN pays Rs 6,000 a year in three equal instalments of Rs 2,000; it took effect from 1 December 2018 and was formally launched in February 2019.
- 22. (C) Suratgarh, Rajasthan** — Prime Minister Narendra Modi launched the Soil Health Card Scheme on 19 February 2015 at Suratgarh in Sriganganagar district of Rajasthan.
- 23. (B) Sikkim** — Sikkim was declared India's first fully organic state in 2016 and won the FAO's Future Policy Gold Award for it.
- 24. (D) I, II and III** — All three are correct. In areas of lower rainfall, such as Punjab and Haryana, rice is grown with canal and tube-well irrigation.
- 25. (A) I and II only** — Statement III is wrong — IR-8 is a rice variety developed at the International Rice Research Institute in the Philippines; the wheat varieties came from Mexico.
- 26. (B) I and III only** — Black soil is rich in lime, iron, magnesia and potash but poor in nitrogen, phosphorus and humus, so statement II is wrong.
- 27. (B) I, II and III** — All three are correct. A2+FL covers paid-out costs plus the imputed value of family labour.

- 28. (D) I, II and III** — All three are correct; PMFBY started in the kharif season of 2016 and enrolment was made voluntary for loanee farmers in 2020.
- 29. (A) I and II only** — India is the second largest producer of wheat after China, so statement III is wrong; India also ranks second in rice, sugarcane and groundnut.
- 30. (C) I and II only** — Gram and mustard are rabi crops, so statement III is wrong; zaid also includes fodder crops and summer moong.
- 31. (A) A-1, B-2, C-3, D-4** — Maharashtra leads in jowar, Uttar Pradesh in wheat, Gujarat in groundnut and Karnataka in mulberry silk; Rajasthan leads in barley, bajra and mustard.
- 32. (C) A-1, B-2, C-3, D-4** — The Yellow Revolution followed the Technology Mission on Oilseeds (1986); the White Revolution relates to milk and the Pink Revolution to prawns and meat processing.
- 33. (D) A-1, B-2, C-3, D-4** — Laterite occurs in the high-rainfall hills of the Western Ghats, Odisha and the north-east; arid sandy soils dominate western Rajasthan.
- 34. (B) A-1, B-2, C-3, D-4** — Walra is the slash-and-burn cultivation practised by the Bhils of southern Rajasthan; other regional names include bewar and dahiya in Madhya Pradesh.
- 35. (C) A-1, B-2, C-3, D-4** — The rice institute at Cuttack was formerly the Central Rice Research Institute; wheat and barley research is centred at Karnal.
- 36. (D) Coffee — Assam** — Coffee is grown in the hills of Karnataka, Kerala and Tamil Nadu; Assam is the leading tea producer. Saffron is grown at Pampore in Kashmir.
- 37. (C) Barley** — Barley is a rabi crop sown in winter; jowar, bajra and cotton are sown with the south-west monsoon.
- 38. (B) It is rich in humus and nitrogen** — Leaching removes lime and silica and washes away organic matter, leaving laterite poor in humus, nitrogen, phosphate and calcium.
- 39. (D) National Food Security Mission — 2016** — The National Food Security Mission was launched in 2007-08 to raise the output of rice, wheat and pulses; the other three pairs are correct.
- 40. (A) I, II, III, IV** — Operation Flood began in 1970, NABARD was set up in 1982, the National Food Security Mission in 2007-08 and PMFBY in 2016.

IG06 — EXPLANATIONS

- 1. (C) Odisha** — Odisha accounts for over half of India's iron ore output, mainly haematite from Keonjhar, Sundargarh and Mayurbhanj districts.
- 2. (A) Chhattisgarh** — Bailadila lies in Dantewada district of Chhattisgarh in the Durg-Bastar-Chandrapur belt; its high-grade haematite is exported to Japan through Visakhapatnam port.
- 3. (B) Madhya Pradesh** — Madhya Pradesh accounts for about 31% of manganese ore production, ahead of Maharashtra and Odisha; Odisha, however, holds the largest reserves of manganese in the country.
- 4. (A) Madhya Pradesh and Maharashtra** — Balaghat is in Madhya Pradesh and Nagpur-Bhandara in Maharashtra; together they form India's most productive manganese belt. Manganese is used mainly in making steel and dry batteries.
- 5. (C) Madhya Pradesh** — Malanjkhand in Balaghat district makes Madhya Pradesh the largest copper-producing state; Khetri in Rajasthan and Ghatsila in Jharkhand are the other Hindustan Copper units.
- 6. (D) Odisha** — Panchpatmali in Koraput district of Odisha feeds NALCO's alumina refinery at Damanjodi and smelter at Angul; Odisha is the leading bauxite producer.
- 7. (A) Rajasthan** — Rampura Agucha in Bhilwara district, along with Zawar and Rajpura-Dariba, makes Rajasthan the source of almost all of India's lead and zinc.
- 8. (B) even numbers increasing from east to west** — North-south highways carry even numbers rising from east to west (NH-44, NH-48, NH-66) and east-west highways carry odd numbers rising from north to south (NH-19, NH-27).
- 9. (D) Raniganj** — Coal mining in India started at Raniganj in West Bengal in 1774; Jharia in Jharkhand is the largest coalfield and the main source of coking coal.

- 10. (D) prime coking coal** — Jharia supplies most of India's prime coking coal, essential for metallurgical use in the steel industry; it lies in the Damodar valley in Jharkhand.
- 11. (C) Kapurdi-Jalipa** — Kapurdi and Jalipa in Barmer district feed the Barmer lignite-based thermal plant; Neyveli (Tamil Nadu) is India's largest lignite field, Panandhro is in Gujarat and Talcher in Odisha is a coalfield.
- 12. (B) Barmer basin** — Mangala in the Barmer basin was discovered in 2004 and has made Rajasthan the largest onshore crude oil producing state; the HPCL Rajasthan Refinery at Pachpadra was commissioned in 2026.
- 13. (C) Mumbai High** — Mumbai High, about 176 km off the Maharashtra coast, was discovered in 1974 and drilled by the rig Sagar Samrat; it remains India's largest producing field.
- 14. (A) Digboi** — The Digboi refinery in Assam is the oldest in India and among the oldest in the world; Jamnagar in Gujarat is today the world's largest refining complex.
- 15. (D) Tarapur** — Tarapur in Maharashtra began operating in 1969 with boiling water reactors built with American help; Rawatbhata in Rajasthan followed with Canadian collaboration.
- 16. (A) Kakrapar** — Unit 3 of the Kakrapar Atomic Power Station in Gujarat is the first indigenous 700 MWe PHWR; Kudankulam in Tamil Nadu, built with Russian help, is India's largest nuclear station by capacity.
- 17. (D) West Germany** — Rourkela (Odisha) was built with West German collaboration, Bhilai and Bokaro with Soviet help and Durgapur with British help.
- 18. (B) Mumbai** — The first successful cotton mill was established at Mumbai in 1854; Ahmedabad, which followed in 1861, came to be called the 'Manchester of India'.
- 19. (D) Chennai** — Chennai and its neighbourhood host a large share of India's car and commercial vehicle output; other hubs are Gurugram-Manesar, Pune-Chakan, Sanand and Pithampur.
- 20. (A) Delhi, Kolkata, Chennai and Mumbai** — The Golden Quadrilateral was taken up as Phase-I of the National Highways Development Project and was completed in 2012.
- 21. (B) Jhansi** — The Srinagar-Kanniyakumari corridor (NH-44) and the Silchar-Porbandar corridor (NH-27) cross each other at Jhansi in Uttar Pradesh.
- 22. (A) Silchar and Porbandar** — The East-West Corridor runs from Silchar in Assam to Porbandar in Gujarat, largely along NH-27; the North-South Corridor runs from Srinagar to Kanniyakumari.
- 23. (C) 5,846 km** — The Golden Quadrilateral is about 5,846 km long; its Kolkata-Chennai arm is the longest side and Chennai-Mumbai the shortest.
- 24. (B) NH-44** — NH-44, running from Srinagar to Kanniyakumari, is about 4,113 km long; NH-27 from Porbandar to Silchar is the second longest.
- 25. (D) I, II and III** — All three are correct; the North-South Corridor also has a spur from Salem to Kochi, and the two corridors meet at Jhansi.
- 26. (D) I, II and III** — All three are correct; the national highway network stood at about 1.46 lakh km in 2024-25 and the NHAI became operational in 1995.
- 27. (C) I and II only** — The South Coast Railway zone, functional from 1 June 2026, has its headquarters at Visakhapatnam, so statement III is wrong; with it Indian Railways has 19 zones.
- 28. (D) I, II and III** — All three are correct. India has 12 major ports under the Centre, and Vadhavan in Palghar district was cleared in 2024.
- 29. (B) I and III only** — Jharkhand holds the largest coal reserves, so statement II is wrong; Rajasthan's fossil fuel wealth is lignite, not coal.
- 30. (A) I and II only** — The IWAI, set up in 1986, has its headquarters at Noida, so statement III is wrong; the National Waterways Act, 2016 declared 111 national waterways.
- 31. (A) I and II only** — The Kollam-Thiruvananthapuram region rests on plantation produce and agricultural raw materials — coir, rubber, cashew, sugar and aluminium — not iron and steel.
- 32. (D) I, II and III** — All three are correct; the Western DFC (about 1,506 km) was built with Japanese assistance and the Eastern DFC (about 1,337 km) with World Bank support.
- 33. (A) A-1, B-2, C-3, D-4** — Delhi-Kolkata is NH-19 (old NH-2), Kolkata-Chennai NH-16 (old NH-5), Chennai-Mumbai NH-48 (old NH-4) and the North-South Corridor NH-44.

- 34. (C) A-1, B-2, C-3, D-4** — Jhamarkotra (Udaipur) is India's chief rock phosphate mine, Sukinda in Odisha supplies most of its chromite, Panna in Madhya Pradesh its diamonds and Jaduguda in Jharkhand was its first uranium mine.
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- 35. (D) A-1, B-2, C-3, D-4** — All four are public sector SAIL plants; Visakhapatnam in Andhra Pradesh is the only shore-based integrated steel plant.
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- 36. (C) A-1, B-2, C-3, D-4** — Deendayal is the renamed Kandla port in Gujarat and V. O. Chidambaranar the renamed Tuticorin port; Mormugao handles iron ore exports.
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- 37. (A) A-1, B-2, C-3, D-4** — The Hugli region grew around jute mills, Chotanagpur around coal and iron ore, Gujarat around cotton and petrochemicals, and Vishakhapatnam around shipbuilding and petroleum refining.
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- 38. (A) Bauxite — Punjab** — Punjab has virtually no metallic minerals; bauxite comes mainly from Odisha, Jharkhand, Gujarat and Chhattisgarh.
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- 39. (B) Neyveli** — Neyveli in Tamil Nadu is a lignite-based thermal power complex, not a nuclear station; Kaiga is in Karnataka, Narora in Uttar Pradesh and Kakrapar in Gujarat.
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- 40. (C) NH-48 — Delhi to Kolkata** — NH-48 runs from Delhi to Chennai through Mumbai; the Delhi-Kolkata highway of the Golden Quadrilateral is NH-19.
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- 41. (D) Mundra** — Mundra in Gujarat is a privately developed non-major port and the largest privately operated port of the country; Kamarajar (Ennore) was the first corporatised major port.
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- 42. (A) Chotanagpur — Chhattisgarh** — The Chotanagpur industrial region lies mainly in Jharkhand with extensions into Odisha and West Bengal.
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- 43. (B) Bokaro — West Bengal** — The Bokaro coalfield lies in Jharkhand in the Damodar valley; Raniganj is the major coalfield of West Bengal.
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- 44. (C) I, II, III, IV** — The Tata plant at Sakchi (Jamshedpur) was set up in 1907 and began production in 1912, Bhilai started in 1959, Bokaro in the early 1970s and Visakhapatnam in 1992.
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- 45. (D) A is false but R is true** — Odisha has the largest manganese reserves (about 34%), but Madhya Pradesh leads in production, followed by Maharashtra and then Odisha.
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